

Financial News Sentimental Analysis and their Correlation with Daily Trade Volume

Final project for INST447, Spring 2026

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Executive Summary

This project is designed for investors who hold positions in publicly traded stocks, particularly focused on Apple (AAPL). It may be hard to consistently monitor the constant stream of financial news, analyst opinions, and quantitative commentary that surrounds their investments. Rather than tracking prices, which investors can observe directly, our tool focuses on the sentiment landscape of the news and public conversations around the stock, giving the users a quick pulse on whether the coverage is heading towards positive, negative, or neutral and what stocks lead in that direction. The primary data source is the Finnhub financial news API, which looks at articles from major outlets including Yahoo Finance, Benzinga, SeekingAlpha, CNBC, ChartMill, and Finnhub's own feed. The trade volume data is pulled from Yahoo Finance from the yfinance library. The trade volume is used to compare if positive or negative sentiment has an impact on the trade volume of AAPL.

The key findings were from the 150+ news articles over May 1–2, 2026 using the FinBERT model; it skewed meaningfully positive, driven by Apple's blowout fiscal Q2 2026 earnings report, which recorded a revenue of \$111.2B. This was caused by a 22% surge in iPhone sales, a \$100B share buyback, and beating Wall Street estimates. Neutral was the second most common label, showing market context articles that mentioned AAPL without taking a specific stance. Negative sentiment was a minority but clustered around real concerns, such as the "RAMageddon", the supply constraints of Mac items, and geopolitical tariff risks. A notable bias finding was source concentration, Yahoo Finance dominated the dataset by volume, which means the way they edit their articles largely shapes the overall sentiment distribution. FinBERT confidence was generally high (median above 0.87), with lower confidence observed on mixed or ambiguous articles, validating the model's calibration on financial text. Along with that there is strong evidence of the news sentiment following the overall trend of the stock. Meaning negative sentiment means overall less trades, and high sentiment pushes for higher trade volume. As AAPL is an overall bullish stock, the high confidence of traders means that negative sentiment usually causes hesitation and not a sell off.

The most technically challenging aspect was trying to integrate the FinBert pipeline to work with the full ETL workflow. It was really difficult to manage the limits, and cleaning the characters, and handling API summaries that cut off mid sentence. It was also hard to get started, trying to find a stock that had a multitude of sources that weren't all biased was something that took time to compile. It was also very interesting to determine how to handle articles that mentioned AAPL broadly, especially since they were more focused on the market rather than Apple specifically. The source bias analysis really highlighted how diverse the dataset is yet still so concentrated with so much data.

Project library imports:

```
import requests
from dotenv import load_dotenv
import yfinance as yf
import os
from transformers import pipeline
from datetime import datetime, timedelta
import time
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

Load environment variables:

```
load_dotenv()
```

True

Source(s) & Extraction

We are extracting data from two main sources: yFinance (<https://ranaroussi.github.io/yfinance/>) and Finnhub Market News API (<https://finnhub.io/docs/api/market-news>). Yahoo! finance data is extracted directly from the package's underlying requests to Yahoo's api. You can view the data without the API from Yahoo Finance (example with AAPL - <https://finance.yahoo.com/quote/AAPL/>). Finnhub News API obtained their data through either contracts with news companies, or they obtained it through public databases and scraping it themselves. But Looking at their professionalism and accuracy based on the timestamps, I believe they are in some sort of agreement with all the major news sources therefore allowing them to use their news metadata and sell them. The headlines themselves are generated by authors from those respective news outlets. The summaries however, may have a chance of being generated by AI. Yahoo Finance data are free for anyone and is not accurate in the sense

that it does not show both live data and sub-minute ticks. Therefore it leads me to believe that Yahoo Finance also has a deal with major exchanges to display this data for free, to the public. Since it is minute-ticks, this means that the provider, or Yahoo themselves calculated high, low, volume and other fields values to be displayed.

First the variable of note for yFinance is the 'Volume' since we are trying to compare whether news sentiment has an impact on the trade volume of that specific stock.

Example of the data obtained from yFinance `ticker.history(start=start_date, end=end_date)`:

Table 1: Sample row returned by yFinance for AAPL

Date	Open	High	Low	Close	Volume	Divi- dends	Stock Splits
2026-05-01 00:00:00-04:00	278.60329	286.95561	278.11375	279.88214	179915400	0.0	0.0

This is a sample of what we obtained as a return from yFinance. We specifically will be using the 'Volume' column.

Variable name - Volume (integer) The number of trades conducted on that day is related to that specific ticker.

As for Finnhub, we extracted news data from their api and it returns the following:

```
[
  {
    'category': 'company',
    'datetime': 1622505600,
    'headline': 'Apple announces new iPhone',
    'id': 123456,
    'image': 'https://example.com/image.jpg',
    'related': 'AAPL',
    'source': 'Reuters',
    'summary': 'Apple has announced the release of its new iPhone model, which features a large screen and a new design.',
    'url': 'https://www.reuters.com/article/us-apple-iphone-idUSKCN2D90XG'
  },
]
```

```
[{'category': 'company',
  'datetime': 1622505600,
  'headline': 'Apple announces new iPhone',
```

```
'id': 123456,
'image': 'https://example.com/image.jpg',
'related': 'AAPL',
'source': 'Reuters',
'summary': 'Apple has announced the release of its new iPhone model, which features a large
'url': 'https://www.reuters.com/article/us-apple-iphone-idUSKCN2D90XG'}]]
```

This is a sample of what is returned when we called:

```
"https://finnhub.io/api/v1/company-news?symbol={ticker}&from={start_date}&to={end_date}&token"
```

```
'https://finnhub.io/api/v1/company-news?symbol={ticker}&from={start_date}&to={end_date}&token'
```

We will be using all of these fields for analysis later on, but the main three variables we will be using for our sentiment analysis is the following: ‘datetime’ (unix timestamp - integer) to determine when the article is released. This is crucial because we will eventually be conducting multi-day analysis and we need to separate articles by date. ‘headline’ (string) and ‘summary’ (string) are both used to construct ‘text’ in a later data structure which is used in the sentiment analysis engine. We combine both ‘headline’ and ‘summary’ with a ‘;’ as the separator to form ‘text’(string). This is because finBERT takes a string of text as input and we want to capture both the summary and the headline as input, without losing any context.

The other variables are used in later supplemental analysis. The main one is ‘source’ which we will compare which source is the most bullish or bearish and who is reporting on that specific ticker.

Variable name - “datetime” (integer) The unix timestamp of when the article came out.

Variable name - “headline”(string) The title of the article.

Variable name - “summary” (string) The description of the contents of the article.

Variable name - “source” (string) The name of the news outlet that wrote the article.

Obtaining and combining data from two separate data sources in preparation for analysis is one of the hardest parts of the project. We first obtain the volume data from yFinance using the `get_volume_data`.

```
def get_volume_data(ticker, start_date, end_date):
    """Fetches the volume of trades conducted on the specified company and date.

    Args:
        ticker (str): Stock ticker symbol
        start_date (str): Start date in the format YYYY-MM-DD
```

```

        end_date (str): End date in the format YYYY-MM-DD

Returns:
    pandas.Series: A series containing the volume of trades for the specified company and
    """
    ticker = yf.Ticker(ticker)
    hist = ticker.history(start=start_date, end=end_date)
    return hist["Volume"]

```

This function obtains and returns the trade volume of a given stock for the given time duration. The ticker (example: AAPL) is passed into the yf.Ticker class to obtain the ticker specific information. We then access the history by calling the history() method from the Ticker class, passing in a start and end date to get the volume data from that range.

```

def get_news_company(ticker, start_date, end_date):
    """Fetches news articles from Finnhub for a specific company within a given date range.

    Args:
        ticker (str): Stock ticker symbol
        start_date (str): Start date in the format YYYY-MM-DD
        end_date (str): End date in the format YYYY-MM-DD

    Returns:
        list: A list of news articles for the specified company and date range in this format:
        [
            {
                'category': 'company',
                'datetime': 1622505600,
                'headline': 'Apple announces new iPhone',
                'id': 123456,
                'image': 'https://example.com/image.jpg',
                'related': 'AAPL',
                'source': 'Reuters',
                'summary': 'Apple has announced the release of its new iPhone model, which f
                'url': 'https://www.reuters.com/article/us-apple-iphone-idUSKCN2D90XG'
            }
        ]
    """
    api_key = os.getenv("FINNHUB_API_KEY")
    url = f"https://finnhub.io/api/v1/company-news?symbol={ticker}&from={start_date}&to={end_date}"
    r = requests.get(url)

```

```

if r.status_code == 200:
    return r.json()
else:
    print(f"Error fetching news data: {r.status_code}")
    return None

```

The function above fetches the company headline and summary using the Finnhub API when given a specific ticker symbol, a start and end date. The api requires an api key which is imported with the python_dotenv package. It uses the request function from the requests package to conduct the api request. It also has simple error handling for any unexpected behaviors. The description of the function also provided an example of the return values.

```

def get_news_range(symbol, start_date, end_date):
    """Get the news based on the range of the dates. Increments a day at a time to bypass re

    Args:
        symbol (str): The symbol you are trying to fetch
        start_date (str): Start date in %Y-%m-%d
        end_date (str): End date in %Y-%m-%d

    Returns:
        [dict]: List of all the article metadata
    """
    start = datetime.strptime(start_date, "%Y-%m-%d").date()
    end = datetime.strptime(end_date, "%Y-%m-%d").date()

    all_news = []
    cur = start
    print("Begin fetching news articles:")
    while cur < end:
        day_str = cur.strftime("%Y-%m-%d")
        chunk = get_news_company(symbol, day_str, day_str) or []
        all_news.extend(chunk)
        cur += timedelta(days=1)
        time.sleep(0.05)
        print("Day:", cur)

    return all_news

```

get_news_range extends on the get_news_company function to obtain news articles that are more than a day. This is important because for financial analysis we usually only see the effect over a long period of time, therefore we would require the fetching of a range of dates. Finnhub

also has a limit to how many news articles it could send, and for long range dates, it often silently drops those dates.

```
def data_transformation_engine(news_raw, volume):
    """Combines both yFinance volume data and Finnhub article data.
    Finnhub articles use unix time, which needs to be converted to string time.
    Also converts the Series integer data structure in pandas to python primitive int.
    Iterates through both news and volume data inputs and combines them into a single dataset.

    Args:
        news_raw (list(dict)): A giant list of dictionaries containing metadata information :
        volume (pandas.Series): A pandas series of dates and the traded volumes

    Returns:
        _type_:

        type dataset:
            key: date strings in the format of "%Y-%m-%d"
            value: Day

        type Day:
            'articles': [Article] --> list of articles
            'volume': int

        type Article:
            "ticker": str,
            "headline": str,
            "summary": str,
            "text": str,
            "source": str,
            "date": datetime
    """
    dataset = {}
    news = transform_news_data(news_raw)

    vol_by_date = {}
    for ts, vol in volume.items():
        date_str = ts.strftime("%Y-%m-%d")
        vol_by_date[date_str] = int(vol)

    for article in news:
        date = article['date']
        if date not in vol_by_date:
```

```

        continue
    if date not in dataset:
        dataset[date] = {
            'articles': [article],
            'volume': vol_by_date[date],
        }
    else:
        dataset[date]['articles'].append(article)

# print("Verification:")
# for d in dataset:
#     print(d, len(dataset[d]['articles']), dataset[d]['volume'])

return dataset

```

The snippet above is code from the `multiday_financial_news_sentiment_pipeline` where the extraction code is obtained and combined with the data transformation engine. The `transform_news_data` returns the cleaned, and processed article information containing the ‘text’ field which is the combination of ‘headline’ and ‘summary’ fields.

```

vol_by_date = {}
for ts, vol in volume.items():
    date_str = ts.strftime("%Y-%m-%d")
    vol_by_date[date_str] = int(vol)

```

This snippet changes the string time to the proper format so it could be the same as the volume dates.

```

for article in news:
    date = article['date']
    if date not in vol_by_date:
        continue
    if date not in dataset:
        dataset[date] = {
            'articles': [article],
            'volume': vol_by_date[date],
        }
    else:
        dataset[date]['articles'].append(article)

```

Next, we generate the “day” data structure by constructing a dictionary that has two fields:


```
dataset[date] = {
    'articles': [article],
    'volume': vol_by_date[date],
}
```

This is the basic structure of a “day”. It contains a list of all the articles, and has the volume for the day. The ‘date’ serves as the primary key that connects the two datasets following the format “%Y-%m-%d”.

Data_transformation_engine function takes in inputs from both get_news_company and get_volume_data formats and then merges them into a single data structure. The data is finally returned as a big structure like the following:

```
type dataset:
    key: date strings in the format of "%Y-%m-%d"
    value: Day

type Day:
    'articles': [Article] --> list of articles
    'volume': int

type Article:
    "ticker": str,
    "headline": str,
    "summary": str,
    "text": str,
    "source": str,
    "date": datetime
```

The main results of the merging of the code returns a list of “days”. This is so that the analysis could iterate over this later for visualization and ease of calculation. Visualizations use daily data as the bars in the bar chart, so having each of them in lists assists in decreasing the complexity later down the line.

Transformation Processes

```
def transform_news_data(news_data):

    transformed = []
```

```

for article in news_data:

    headline = article.get("headline", "")
    summary = article.get("summary", "")

    # combine text
    text = f"{headline}. {summary}"

    # cleaning
    text = text.replace("\xa0", " ")
    text = text.replace("&#39;", "'")
    text = " ".join(text.split())
    text = text.encode("utf-8", "ignore").decode("utf-8")

    # convert timestamp
    date = datetime.fromtimestamp(
        article["datetime"]
    ).strftime("%Y-%m-%d")

    transformed.append({
        "ticker": article.get("related"),
        "headline": headline,
        "summary": summary,
        "text": text,
        "source": article.get("source"),
        "date": date
    })

return transformed

def get_news_range(symbol, start_date, end_date):
    """Get the news based on the range of the dates. Increments a day at a time to bypass retr

    Args:
        symbol (str): The symbol you are trying to fetch
        start_date (str): Start date in %Y-%m-%d
        end_date (str): End date in %Y-%m-%d

    Returns:
        [dict]: List of all the article metadata
    """
    start = datetime.strptime(start_date, "%Y-%m-%d").date()

```

```

end = datetime.strptime(end_date, "%Y-%m-%d").date()

all_news = []
cur = start
print("Begin fetching news articles:")
while cur < end:
    day_str = cur.strftime("%Y-%m-%d")
    chunk = get_news_company(symbol, day_str, day_str) or []
    all_news.extend(chunk)
    cur += timedelta(days=1)
    time.sleep(0.05)
    print("Day:", cur)

return all_news

def data_transformation_engine(news_raw, volume):
    """Combines both yFinance volume data and Finnhub article data.
    Finnhub articles uses unix time, which needs to be converted to string time.
    Also converts the Series integer data structure in pandas to python primitive int.
    Iterates through both news and volume data inputs and combines them into a single dataset

    Args:
        news_raw (list(dict)): A giant list of dictionaries containing metadata information f
        volume (pandas.Series): A pandas series of dates and the traded volumes

    Returns:
        _type_:

        type dataset:
            key: date strings in the format of "%Y-%m-%d"
            value: Day

        type Day:
            'articles': [Article] --> list of articles
            'volume': int

        type Article:
            "ticker": str,
            "headline": str,
            "summary": str,
            "text": str,
            "source": str,

```

```

        "date": datetime
    """
    dataset = {}
    news = transform_news_data(news_raw)

    vol_by_date = {}
    for ts, vol in volume.items():
        date_str = ts.strftime("%Y-%m-%d")
        vol_by_date[date_str] = int(vol)

    for article in news:
        date = article['date']
        if date not in vol_by_date:
            continue
        if date not in dataset:
            dataset[date] = {
                'articles': [article],
                'volume': vol_by_date[date],
            }
        else:
            dataset[date]['articles'].append(article)

    # print("Verification:")
    # for d in dataset:
    #     print(d, len(dataset[d]['articles']), dataset[d]['volume'])

    return dataset


def sentimental_analysis_engine(sen_engine, dataset):

    for day in dataset:
        txt_buffer = [txt['text'] for txt in dataset[day]['articles']]
        dataset[day]['sentiment_results'] = sen_engine(txt_buffer)


def multiday_financial_news_sentiment_pipeline(symbol, start_date, end_date, sen_engine):
    """Multiday news sentiment engine. Uses FinBERT as the sentimental analysis. Obtains daily

    Args:
        symbol (str): The symbol you are trying to fetch
        start_date (str): Start date in %Y-%m-%d

```

```

    end_date (str): End date in %Y-%m-%d

Returns:
    variable: transformed_dataset

    type dict Transformed_Dataset:
        key: date strings in the format of "%Y-%m-%d"
        value: type Day

    variable: final_dataset:

    type list Final_Dataset:
        items: type Article

    type dict Day:
        'articles': [Article] --> list of articles
        'volume': int

    type dict Article:
        "ticker": str,
        "headline": str,
        "summary": str,
        "text": str,
        "source": str,
        "date": datetime,
        "sentiment": int,
        "sen_confidence": int

    variable:

"""
# Extraction
news_raw = get_news_range(symbol, start_date, end_date)
vol = get_volume_data(symbol, start_date, end_date)

# Transformation
transformed_dataset = data_transformation_engine(news_raw, vol)

# Load
sentimental_analysis_engine(sen_engine, transformed_dataset)

# Build final dataset

```

```

# print(clean_data.keys()) --> Days
# print(clean_data['2026-05-01'].keys()) --> {'articles', 'volume', 'sentiment_results'}

final_dataset = []

for day in transformed_dataset:
    for article, sentiment in zip(transformed_dataset[day]['articles'], transformed_data

        article["sentiment"] = sentiment["label"]
        article["sen_confidence"] = sentiment["score"]

        final_dataset.append(article)
        # print(article.keys()) --> {'ticker', 'headline', 'summary', 'text', 'source', '

return [final_dataset, transformed_dataset]

if __name__ == "__main__":

    # Load finbert model
    finbert = pipeline(
        "text-classification",
        model="ProsusAI/finbert",
        tokenizer="ProsusAI/finbert"
    )

    # Set start and end dates, in our case we will use May 01 - May 02
    start_date = "2026-05-01"
    end_date = "2026-05-08"
    symbol = "AAPL"

    # Obtain news regarding a specific symbol (in our case it will be Apple)
    [final_dataset, results] = multiday_financial_news_sentiment_pipeline(symbol, start_date

    # OUTPUT
    df = pd.DataFrame(final_dataset)
    df["date"] = pd.to_datetime(df["date"])

    for row in final_dataset:
        print("\n-----")

```

```
print(f"Headline: {row['headline']}")
print(f"Sentiment: {row['sentiment']}")
print(f"Confidence: {row['sen_confidence']:.2f}")
print(f>Date: {row['date']}")
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable authentication.

Loading weights: 0% | 0/201 [00:00<?, ?it/s]

Begin fetching news articles:

Day: 2026-05-02

Day: 2026-05-03

Day: 2026-05-04

Day: 2026-05-05

Day: 2026-05-06

Day: 2026-05-07

Day: 2026-05-08

Headline: At Risk in North American Trade Talks: Cheap Cars
Sentiment: negative
Confidence: 0.50
Date: 2026-05-01

Headline: Replit's Amjad Masad on the Cursor deal, fighting Apple, and why he'd rather not s
Sentiment: neutral
Confidence: 0.89
Date: 2026-05-01

Headline: Stock Market Today, May 1: Apple Jumps After Record Quarter and \$100 Billion Share
Sentiment: positive
Confidence: 0.88
Date: 2026-05-01

Headline: S&P 500, Nasdaq end higher, notch weekly gains
Sentiment: positive
Confidence: 0.87
Date: 2026-05-01

Headline: Netflix Bets Big On Theaters With Greta Gerwig's 'Narnia: The Magician's Nephew' In
Ever Wide Theatrical Rollout
Sentiment: positive
Confidence: 0.87
Date: 2026-05-01

Headline: Stock Market Today, May 1: S&P 500 and Nasdaq Power to New Highs
Sentiment: positive
Confidence: 0.93
Date: 2026-05-01

Headline: Dow Jones Futures: Stock Market Hits Highs After Bullish Pause; Apple, Broadcom In
Sentiment: positive
Confidence: 0.66
Date: 2026-05-01

Headline: Nasdaq, S&P 500 Log Record Finish, Extend Streak of Weekly Gains
Sentiment: positive
Confidence: 0.90
Date: 2026-05-01

Headline: US Equity Indexes Close Mixed as Earnings Lift Technology While Trump 'Not Happy' W
Sentiment: negative
Confidence: 0.55
Date: 2026-05-01

Headline: S&P 500, Nasdaq end at fresh records on tech earnings strength
Sentiment: positive
Confidence: 0.49
Date: 2026-05-01

Headline: Big Tech powers the market with strong earnings wins
Sentiment: neutral
Confidence: 0.80
Date: 2026-05-01

Headline: S&P 500 and Nasdaq 100 Post Record Highs on Solid Earnings
Sentiment: positive
Confidence: 0.90
Date: 2026-05-01

Headline: Stock Market Today: Nasdaq, S&P Make History; Buffett's Berkshire Reports Earnings
Sentiment: neutral
Confidence: 0.94
Date: 2026-05-01

Headline: Why Apple Stock is Jumping on Friday
Sentiment: positive
Confidence: 0.92
Date: 2026-05-01

Headline: Nasdaq Smashes 25,000 Barrier in Historic First
Sentiment: positive
Confidence: 0.89
Date: 2026-05-01

Headline: How major US stock indexes fared Friday 5/1/2026
Sentiment: negative
Confidence: 0.94
Date: 2026-05-01

Headline: US Equity Indexes Mixed in Final Leg of Trading as Trump 'Not Happy' With Iran's R
Sentiment: negative
Confidence: 0.97
Date: 2026-05-01

Headline: Apple Stock Surged After Earnings. 3 Concerns Lurking Beneath the Surface.
Sentiment: positive
Confidence: 0.84
Date: 2026-05-01

Headline: 3 Stocks That Could Be Your Easy Ticket to Creating Generational Wealth
Sentiment: positive
Confidence: 0.63
Date: 2026-05-01

Headline: Apple Warns the Memory Shortage Is Getting Worse. Experts Say It Could Drive Up Pr
Sentiment: negative
Confidence: 0.96
Date: 2026-05-01

Headline: Dan Ives: Tech stocks have 12-15% more upside - his top AI buys now
Sentiment: neutral
Confidence: 0.87
Date: 2026-05-01

Headline: 3 Takeaways From Apple Earnings. The Stock Is Surging.
Sentiment: positive
Confidence: 0.92
Date: 2026-05-01

Headline: Equity Markets Mostly Rise Intraday as Apple Helps Lift Tech
Sentiment: positive
Confidence: 0.94
Date: 2026-05-01

Headline: Friday's session: most active stock in the S&P500 index
Sentiment: neutral
Confidence: 0.88
Date: 2026-05-01

Headline: Dividend Champion, Contender, And Challenger Highlights: Week Of May 3
Sentiment: neutral
Confidence: 0.94
Date: 2026-05-01

Headline: Friday's session: most active stocks
Sentiment: neutral
Confidence: 0.89
Date: 2026-05-01

Headline: Why the S&P 500 and Nasdaq Hit Record Highs While the Dow Fell Today
Sentiment: neutral
Confidence: 0.87
Date: 2026-05-01

Headline: Technology, Consumer Discretionary Push S&P 500, Nasdaq to Record Highs
Sentiment: positive
Confidence: 0.94
Date: 2026-05-01

Headline: Apple's Strong Revenue Guidance Came Despite Mac Supply Constraints, Wedbush Says
Sentiment: positive
Confidence: 0.95
Date: 2026-05-01

Headline: Tech Up After Apple Earnings -- Tech Roundup
Sentiment: positive
Confidence: 0.95
Date: 2026-05-01

Headline: Apple Well Positioned for Continued Growth After Strong Fiscal Q2, BofA Securities
Sentiment: positive
Confidence: 0.95
Date: 2026-05-01

Headline: 'Fast Money' traders recap a week of mega-cap earnings
Sentiment: neutral
Confidence: 0.91
Date: 2026-05-01

Headline: Apple Q2 Earnings Beat Estimates, iPhone Drives Top-Line Growth

Sentiment: positive
Confidence: 0.93
Date: 2026-05-01

Headline: The Week That Was: May 1, 2026
Sentiment: neutral
Confidence: 0.94
Date: 2026-05-01

Headline: UBS Maintains Neutral on Apple, Raises Price Target to \$296
Sentiment: positive
Confidence: 0.52
Date: 2026-05-01

Headline: Trump raises tariffs on EU vehicles
Sentiment: neutral
Confidence: 0.72
Date: 2026-05-01

Headline: Stock Market Today, May 1: Tech Stocks Soar on Apple Earnings
Sentiment: positive
Confidence: 0.94
Date: 2026-05-01

Headline: These S&P500 stocks have an unusual volume in today's session
Sentiment: neutral
Confidence: 0.86
Date: 2026-05-01

Headline: How Long Have You Owned a Stock?
Sentiment: neutral
Confidence: 0.93
Date: 2026-05-01

Headline: Markets Now: May 1, 2026
Sentiment: neutral

Confidence: 0.94
Date: 2026-05-01

Headline: The Post Market Wrap: May 1, 2026
Sentiment: neutral
Confidence: 0.90
Date: 2026-05-01

Headline: Stock Market Today: Nasdaq Leads, Hits New High; Data Center Play Leaps (Live Cover
Sentiment: negative
Confidence: 0.90
Date: 2026-05-01

Headline: Apple's iPhone Supercycle, Strong Outlook Drive Momentum, Wedbush Says
Sentiment: positive
Confidence: 0.93
Date: 2026-05-01

Headline: Meet the Low-Growth Dow Dividend King That's Outperforming Every "Magnificent Seven"
Sentiment: positive
Confidence: 0.66
Date: 2026-05-01

Headline: S&P 500, Nasdaq Scale New Peaks Amid Strong Gains in Technology
Sentiment: positive
Confidence: 0.94
Date: 2026-05-01

Headline: What's going on in today's session: S&P500 movers
Sentiment: neutral
Confidence: 0.89
Date: 2026-05-01

Headline: Prediction: This Will Be the Next \$4 Trillion Company (and It's Not Microsoft)
Sentiment: negative
Confidence: 0.72

Date: 2026-05-01

Headline: Apple's iPhone sales lift stock, but global memory shortage and supply constraints
Sentiment: positive
Confidence: 0.76
Date: 2026-05-01

Headline: Wall Street Splits on Apple After Q2 Beat: Is the Growth Story Real or Just a Tariff
Sentiment: positive
Confidence: 0.89
Date: 2026-05-01

Headline: Apple delivers strong quarter with upbeat outlook
Sentiment: positive
Confidence: 0.95
Date: 2026-05-01

Headline: Top Midday Stories: Apple Tops Fiscal Q2 EPS, Revenue Estimates, Raises Dividend; I
Sentiment: positive
Confidence: 0.92
Date: 2026-05-01

Headline: Stock Market Rallies To New Highs; Apple, Google, Amazon, Sandisk, Powell In Focus
Sentiment: positive
Confidence: 0.55
Date: 2026-05-01

Headline: Apple beats expectations but flags rising costs
Sentiment: positive
Confidence: 0.94
Date: 2026-05-01

Headline: Bam Azizi, Mesh CEO: A Fortt Knox Conversation
Sentiment: neutral
Confidence: 0.90
Date: 2026-05-01

Headline: Stock Market Today: Dow Inches Up With Help From Apple; Reddit Rallies (Live Coverage)
Sentiment: positive
Confidence: 0.94
Date: 2026-05-01

Headline: Apple is nearing its first record close since December after earnings
Sentiment: positive
Confidence: 0.84
Date: 2026-05-01

Headline: Bumped Apple's price target after gross margins were better than expected: Morgan Stanley
Sentiment: neutral
Confidence: 0.80
Date: 2026-05-01

Headline: Apple Pops 5% on Q2 Beat: Has the iPhone Maker Found Its Growth Story Again?
Sentiment: positive
Confidence: 0.94
Date: 2026-05-01

Headline: Apple Can't Meet Demand For iPhones Due To Chip Shortages
Sentiment: negative
Confidence: 0.96
Date: 2026-05-01

Headline: Apple To Rally More Than 19%? Here Are 10 Top Analyst Forecasts For Friday
Sentiment: neutral
Confidence: 0.62
Date: 2026-05-01

Headline: CNBC TechCheck Evening Edition: May 1, 2026
Sentiment: neutral
Confidence: 0.92
Date: 2026-05-01

Headline: TD Cowen Maintains Buy on Apple, Raises Price Target to \$335
Sentiment: positive
Confidence: 0.83
Date: 2026-05-01

Headline: TSX opens lower as energy volatility persists
Sentiment: negative
Confidence: 0.98
Date: 2026-05-01

Headline: The CBOE VIX Falls to 16 Level as Risk-On Trade Returns to Market
Sentiment: negative
Confidence: 0.97
Date: 2026-05-01

Headline: S&P 500 and Nasdaq Start May at Fresh Record Highs
Sentiment: positive
Confidence: 0.91
Date: 2026-05-01

Headline: Apple Delivers The Quarter Bulls Were Waiting For - Stock Reaches For New Highs
Sentiment: positive
Confidence: 0.95
Date: 2026-05-01

Headline: Why Apple remains an underdog story
Sentiment: neutral
Confidence: 0.79
Date: 2026-05-01

Headline: S&P 500 profit eyes sharpest quarterly growth in four years after Big Tech results
Sentiment: positive
Confidence: 0.95
Date: 2026-05-01

Headline: Nasdaq Composite Takes Off as Tech Bulls Retake Control on Strong Earnings, M&A
Sentiment: positive
Confidence: 0.91
Date: 2026-05-01

Headline: Stock Market Today: Dow Rises As Apple Jumps On Earnings; Roblox, Sandisk, Western
Sentiment: positive
Confidence: 0.40
Date: 2026-05-01

Headline: S&P 500, Nasdaq Rise to Start May at Fresh Record Highs
Sentiment: positive
Confidence: 0.90
Date: 2026-05-01

Headline: Apple Proves AI Doesn't Matter
Sentiment: negative
Confidence: 0.73
Date: 2026-05-01

Headline: Apple credits 'most popular' ever iPhone for booming sales
Sentiment: positive
Confidence: 0.95
Date: 2026-05-01

Headline: 10 Information Technology Stocks Whale Activity In Today's Session
Sentiment: neutral
Confidence: 0.63
Date: 2026-05-01

Headline: Wall Street Roundup: Tech Reports
Sentiment: neutral
Confidence: 0.92
Date: 2026-05-01

Headline: Big Tech Is Spending an Unprecedented Amount on AI. Only 1 Company Is Making It Wo

Sentiment: neutral
Confidence: 0.75
Date: 2026-05-01

Headline: Exchange-Traded Funds Higher, Equity Futures Mixed Amid Continued Earnings Results
Sentiment: positive
Confidence: 0.95
Date: 2026-05-01

Headline: Blowout Earnings From SanDisk And Western Digital But Stock Reaction Is A Tell; 'M
Sentiment: neutral
Confidence: 0.83
Date: 2026-05-01

Headline: S&P 500's Strong May Footing Powered by Apple, SanDisk, and Easing Oil
Sentiment: negative
Confidence: 0.88
Date: 2026-05-01

Headline: Stock Market Today: Apple Earnings Power S&P 500, Nasdaq To Records; Trump Hits EU
Sentiment: negative
Confidence: 0.65
Date: 2026-05-01

Headline: Final Trades: Apollo Global, Archer-Daniels-Midland and Digital Realty
Sentiment: neutral
Confidence: 0.95
Date: 2026-05-01

Headline: Watch CNBC's full interview with Berkshire CEO Greg Abel
Sentiment: neutral
Confidence: 0.94
Date: 2026-05-01

Headline: This Tech ETF Gives the Best Bang for Your Buck
Sentiment: positive

Confidence: 0.95
Date: 2026-05-01

Headline: Dow futures rise 200 points as Apple stock climbs, oil falls
Sentiment: positive
Confidence: 0.94
Date: 2026-05-01

Headline: US Equity Futures Nudge Higher Pre-Bell After S&P 500, Nasdaq Reach New Closing High
Sentiment: positive
Confidence: 0.93
Date: 2026-05-01

Headline: Stocks making the biggest moves midday: Atlassian, Apple, Roblox, Reddit, Twilio & Zoom
Sentiment: neutral
Confidence: 0.91
Date: 2026-05-01

Headline: NYSE Content Update: Artemis II Crew to Take Part in 'Bell Moment' at NYSE
Sentiment: neutral
Confidence: 0.95
Date: 2026-05-01

Headline: The Structural Shift Lifting Apple Stock's Valuation
Sentiment: negative
Confidence: 0.82
Date: 2026-05-01

Headline: Yen Resumes Rally Sparked by First Intervention in Two Years
Sentiment: positive
Confidence: 0.55
Date: 2026-05-01

Headline: Trade School: How to trade Intel at these levels
Sentiment: neutral
Confidence: 0.54

Date: 2026-05-01

Headline: The Late Morning Rundown: May 1, 2026

Sentiment: neutral

Confidence: 0.90

Date: 2026-05-01

Headline: Apple stages sharpest rally in 9 months as execs cite iPhone, Mac demand in boosting

Sentiment: positive

Confidence: 0.95

Date: 2026-05-01

Headline: What's going on in today's pre-market session: S&P500 movers

Sentiment: neutral

Confidence: 0.86

Date: 2026-05-01

Headline: Apple Shrugs Off Component Costs To Post Record March Quarter: Analyst

Sentiment: positive

Confidence: 0.94

Date: 2026-05-01

Headline: These Stocks Are Today's Movers: Apple, Sandisk, Reddit, Western Digital, Moderna,

Sentiment: negative

Confidence: 0.97

Date: 2026-05-01

Headline: Apple Forecasts 14% To 17% Growth As Shares Rise Nearly 4%

Sentiment: positive

Confidence: 0.90

Date: 2026-05-01

Headline: Apple Crushes Q2 Estimates on Surging iPhone and China Sales

Sentiment: positive

Confidence: 0.64

Date: 2026-05-01

Headline: Goldman revamps Apple stock price target for rest of 2026
Sentiment: positive
Confidence: 0.72
Date: 2026-05-01

Headline: Apple Beats Estimates, Forecast Tops Expectations on iPhone Demand
Sentiment: positive
Confidence: 0.94
Date: 2026-05-01

Headline: 5 Things to Know Before the Stock Market Opens
Sentiment: neutral
Confidence: 0.93
Date: 2026-05-01

Headline: Jim Cramer on MP Materials: "They've Got the Government's Backing"
Sentiment: neutral
Confidence: 0.88
Date: 2026-05-01

Headline: Wedbush Reiterates Outperform on Apple, Maintains \$350 Price Target
Sentiment: negative
Confidence: 0.78
Date: 2026-05-01

Headline: Apple's earnings beat is part of an amazing trend
Sentiment: positive
Confidence: 0.85
Date: 2026-05-01

Headline: Mizuho's Jordan Klein on new Apple CEO: Investors are looking for new hardware inn
Sentiment: neutral
Confidence: 0.80
Date: 2026-05-01

Headline: Stocks Mostly Up Pre-Bell as Investors Await More Corporate Earnings
Sentiment: positive
Confidence: 0.88
Date: 2026-05-01

Headline: Why OpenAI's Smartphone Will Disrupt Apple, But Alphabet Will Ultimately Win
Sentiment: negative
Confidence: 0.83
Date: 2026-05-01

Headline: Barclays Maintains Underweight on Apple, Raises Price Target to \$253
Sentiment: positive
Confidence: 0.79
Date: 2026-05-01

Headline: Wells Fargo Maintains Overweight on Apple, Raises Price Target to \$310
Sentiment: positive
Confidence: 0.80
Date: 2026-05-01

Headline: Apple CEO warns memory constraints are intensifying
Sentiment: negative
Confidence: 0.95
Date: 2026-05-01

Headline: Social Buzz: Wallstreetbets Stocks Mostly Advancing Premarket Friday; Reddit, Apple
Sentiment: positive
Confidence: 0.92
Date: 2026-05-01

Headline: Apple Delivers: New CEO, Looking Ahead To WWDC
Sentiment: positive
Confidence: 0.88
Date: 2026-05-01

Headline: Roku Rewinds Doubters With Ad-Fueled Earnings Beat
Sentiment: positive
Confidence: 0.94
Date: 2026-05-01

Headline: Warren Buffett Won't Be The Central Figure At Saturday's Berkshire Hathaway Annual
Sentiment: negative
Confidence: 0.55
Date: 2026-05-01

Headline: Apple shares rise on strong quarterly sales in run-up to CEO change
Sentiment: negative
Confidence: 0.59
Date: 2026-05-01

Headline: Porsche will compete Laguna Seca in the historic colours of the Apple Computer live
Sentiment: neutral
Confidence: 0.88
Date: 2026-05-01

Headline: Apple Is Changing The Rules Right Before Tim Cook Exits
Sentiment: positive
Confidence: 0.74
Date: 2026-05-01

Headline: The downstream winners of the AI trade's latest phase
Sentiment: neutral
Confidence: 0.92
Date: 2026-05-01

Headline: Apple Stock Rises After Earnings Beat, Strong Guidance, Buyback, Dividend Hike
Sentiment: positive
Confidence: 0.94
Date: 2026-05-01

Headline: APPLE INC : Jefferies reiterates its Neutral rating

Sentiment: neutral
Confidence: 0.83
Date: 2026-05-01

Headline: Apple outlook; oil holds gains; yen in focus - what's moving markets
Sentiment: negative
Confidence: 0.81
Date: 2026-05-01

Headline: US stocks drift higher and oil prices hold steady as many world markets are closed
Sentiment: positive
Confidence: 0.91
Date: 2026-05-01

Headline: Apple shares rise 3.9% after co forecasts Q3 sales growth above estimates
Sentiment: positive
Confidence: 0.91
Date: 2026-05-01

Headline: Markets Edge Higher as Apple Outlook Lifts Sentiment, Oil Holds Firm: Dow Jones, S
Sentiment: positive
Confidence: 0.92
Date: 2026-05-01

Headline: Apple Shares Edge Higher on Strong Results and "Remarkable" Margin Outlook
Sentiment: positive
Confidence: 0.95
Date: 2026-05-01

Headline: Apple: Great Earnings, Expensive Stock, Same Hold Rating
Sentiment: positive
Confidence: 0.59
Date: 2026-05-01

Headline: Stock Market Today: Dow, S&P 500, Nasdaq Futures RiseAs Trump Administration Evades
Sentiment: positive

Confidence: 0.78
Date: 2026-05-01

Headline: Apple's Record March Quarter Drowns Out Memory Cost Worries: Analyst
Sentiment: negative
Confidence: 0.67
Date: 2026-05-01

Headline: Apple climbs on earnings as China 'now a strong tailwind', says Wedbush
Sentiment: positive
Confidence: 0.96
Date: 2026-05-01

Headline: Rosenblatt Maintains Neutral on Apple, Raises Price Target to \$276
Sentiment: positive
Confidence: 0.60
Date: 2026-05-01

Headline: Tech Earnings Crack Open a Fault Line, Alphabet and Apple Take the High Ground
Sentiment: neutral
Confidence: 0.63
Date: 2026-05-01

Headline: Stock Futures Rise as Strong Earnings Season Rolls On
Sentiment: positive
Confidence: 0.92
Date: 2026-05-01

Headline: Apple: The CapEx-Lite AI Winner The Market Is Ignoring (Rating Upgrade)
Sentiment: negative
Confidence: 0.46
Date: 2026-05-01

Headline: Apple Inc (AAPL) Q2 2026 Earnings Call Highlights: Record Revenue and Strategic Sh
Sentiment: positive
Confidence: 0.95

Date: 2026-05-01

Headline: Market-Moving News for May 1st
Sentiment: positive
Confidence: 0.95
Date: 2026-05-01

Headline: Morgan Stanley Maintains Overweight on Apple, Raises Price Target to \$330
Sentiment: positive
Confidence: 0.78
Date: 2026-05-01

Headline: Needham Reiterates Hold on Apple to Hold
Sentiment: neutral
Confidence: 0.71
Date: 2026-05-01

Headline: Wall Street Breakfast Podcast: Sandisk Beats, Shares Slip
Sentiment: negative
Confidence: 0.93
Date: 2026-05-01

Headline: Apple Earnings: This Important High-Margin Segment Is Seeing Accelerating Momentum
Sentiment: positive
Confidence: 0.95
Date: 2026-05-01

Headline: Unprecedented AI Spending Is Rewriting Big Tech's Earnings Season Playbook
Sentiment: neutral
Confidence: 0.54
Date: 2026-05-01

Headline: S&P 500, Dow futures inch up after Wall Street's best month in years
Sentiment: positive
Confidence: 0.93
Date: 2026-05-01

Headline: Conference Call:

Apple Sees Q3 Sales \$107.201B-\$110.022B vs \$102.927B Est

Sentiment: neutral

Confidence: 0.65

Date: 2026-05-01

Headline: This Tech Sector ETF Gives Investors the Best Bang for Their Buck

Sentiment: positive

Confidence: 0.95

Date: 2026-05-01

Headline: Apple shares rise as iPhone 17 and MacBook Neo drive forecast

Sentiment: positive

Confidence: 0.86

Date: 2026-05-01

Headline: Will S&P 500 Open Up Or Down On May 1 After Best Month Since 2020?

Sentiment: negative

Confidence: 0.65

Date: 2026-05-01

Headline: Stock Futures Higher, Oil Elevated as European and Asian Markets Shut

Sentiment: positive

Confidence: 0.89

Date: 2026-05-01

Headline: Dan Ives Says Apple's 'Golden Era' Is Near, Gene Munster Sees 10% Upside And Danie

Sentiment: positive

Confidence: 0.91

Date: 2026-05-01

Headline: Apple Q2 Results: Big Growth, But Why You Shouldn't Buy

Sentiment: positive

Confidence: 0.83

Date: 2026-05-01

Headline: Amazon, Alphabet, Microsoft, Meta, and Apple Just Reported Earnings. I Think This V
Sentiment: positive
Confidence: 0.74
Date: 2026-05-01

Headline: Why it's so Enticing to Bite on Apple's Stock After Record Q2 Results
Sentiment: positive
Confidence: 0.94
Date: 2026-05-01

Headline: Apple Weighs Major AI Deal As Investors Reassess Growth And Valuation
Sentiment: neutral
Confidence: 0.79
Date: 2026-05-04

Headline: Apple's Tim Cook says AI-fueled 'RAMmageddon' will make your next laptop and phone
Sentiment: neutral
Confidence: 0.78
Date: 2026-05-04

Headline: Brookwood Cuts FLXR Position by \$3 Million -- What Income Investors Should Know
Sentiment: negative
Confidence: 0.80
Date: 2026-05-04

Headline: Warren Buffett's Favorite Stock That's Up 107,400% Since 1990 Is Still a Buy. Here
Sentiment: neutral
Confidence: 0.64
Date: 2026-05-04

Headline: Bond Yields Are Rising. Why That Could End the Stock Market's Rally.
Sentiment: negative
Confidence: 0.94
Date: 2026-05-04

Headline: Sector Update: Tech Stocks Mixed Late Afternoon
Sentiment: negative
Confidence: 0.94
Date: 2026-05-04

Headline: Congresswoman Ditches Magnificent Seven Stocks, Including The Year-To-Date Leader
Sentiment: negative
Confidence: 0.85
Date: 2026-05-04

Headline: More AI Is Coming to iOS 27. Why That Could Make Apple Stock a Buy Here.
Sentiment: neutral
Confidence: 0.74
Date: 2026-05-04

Headline: Final Trade: AAPL, SPY, DIS
Sentiment: neutral
Confidence: 0.94
Date: 2026-05-04

Headline: AI Trade Sends Asian Tech Shares Up 4.7% To Record
Sentiment: positive
Confidence: 0.94
Date: 2026-05-04

Headline: Apple Plans to Let Users Build Their Own Passes in iOS 27 Wallet App
Sentiment: neutral
Confidence: 0.69
Date: 2026-05-04

Headline: Ross Gerber, Dan Ives Bet On The Same 9 Stocks - Some Might Surprise You
Sentiment: neutral
Confidence: 0.92
Date: 2026-05-04

Headline: Is It Too Late To Buy RIOT Stock?
Sentiment: positive
Confidence: 0.62
Date: 2026-05-04

Headline: Which Will Double First: Amazon Or Oracle? Watch This Kentucky Derby To Find Out
Sentiment: neutral
Confidence: 0.84
Date: 2026-05-04

Headline: Morgan Stanley resets Apple stock price target after earnings
Sentiment: positive
Confidence: 0.79
Date: 2026-05-04

Headline: Intel's Next Chapter Government Backing Foundry Ambitions And AI Partnerships
Sentiment: positive
Confidence: 0.90
Date: 2026-05-04

Headline: Panos Panay on Amazon Entering Starlink's Space
Sentiment: neutral
Confidence: 0.89
Date: 2026-05-04

Headline: Actor Bryan Cranston on launching mezcal and tequila brand Dos Hombres
Sentiment: neutral
Confidence: 0.92
Date: 2026-05-04

Headline: Apple Poised for 22% Surge
Sentiment: positive
Confidence: 0.92
Date: 2026-05-04

Headline: Qualcomm Stock To \$340?

Sentiment: negative

Confidence: 0.70

Date: 2026-05-04

Headline: 'Apple Plans to Let Users Build Their Own Passes in iOS 27 Wallet App' - Bloomberg

Sentiment: neutral

Confidence: 0.92

Date: 2026-05-04

Headline: Apple Hits Near-Record High, While Chip Stocks Face Pullback Risk After Massive Ra

Sentiment: negative

Confidence: 0.48

Date: 2026-05-04

Headline: Palantir Earnings Loom; GameStop's eBay Ambitions; China's Defiance Of US Sanctions

Sentiment: neutral

Confidence: 0.79

Date: 2026-05-04

Headline: Did Berkshire's annual meeting feel the same without Buffett?

Sentiment: neutral

Confidence: 0.94

Date: 2026-05-04

Headline: Prediction: Amazon Stock Will Soon Join the \$3 Trillion Club

Sentiment: neutral

Confidence: 0.78

Date: 2026-05-04

Headline: Interpreting Apple (AAPL) International Revenue Trends

Sentiment: neutral

Confidence: 0.94

Date: 2026-05-04

Headline: Investors Heavily Search Apple Inc. (AAPL): Here is What You Need to Know

Sentiment: neutral
Confidence: 0.92
Date: 2026-05-04

Headline: The Fed Could Crush Trump's Bull Market - Here's the Warning Wall Street Is Missing
Sentiment: neutral
Confidence: 0.61
Date: 2026-05-04

Headline: AI Enthusiasm Powers Nasdaq to All-Time High: 3 Stocks to Grab
Sentiment: positive
Confidence: 0.93
Date: 2026-05-04

Headline: Why Apple's Rumored AI Mega-Deal Could Define Tim Cook's Successor
Sentiment: neutral
Confidence: 0.90
Date: 2026-05-04

Headline: Apple Services Strength Offsets Slower App Downloads
Sentiment: positive
Confidence: 0.94
Date: 2026-05-04

Headline: AI Stocks: Winners Shift As Google Leads. Palantir, Arista, CoreWeave Earnings On L
Sentiment: positive
Confidence: 0.56
Date: 2026-05-04

Headline: Nvidia, Apple, Tesla Are Shut Out Of S&P 500's New Ruling Class
Sentiment: negative
Confidence: 0.95
Date: 2026-05-04

Headline: Anchor Investment Exits Nearly All of Its BSCQ Position -- a \$21 Million Trim
Sentiment: neutral

Confidence: 0.93
Date: 2026-05-04

Headline: Apple Just Gave Investors a \$100 Billion Reason to Rethink the AI Spending Race
Sentiment: positive
Confidence: 0.82
Date: 2026-05-04

Headline: Rep. Lizzie Fletcher Has Sold Up to \$165K Worth Of Alphabet Stock: Here's What You
Sentiment: neutral
Confidence: 0.93
Date: 2026-05-04

Headline: Mark Cuban Slams OpenAI For Overspending (CORRECTED)
Sentiment: negative
Confidence: 0.87
Date: 2026-05-04

Headline: DEI Programs Are Dwindling But the Trump Administration Isn't Done
Sentiment: positive
Confidence: 0.52
Date: 2026-05-04

Headline: Tech stocks today: Semiconductor company earnings in focus amid AI boom, Musk vs. J
Sentiment: negative
Confidence: 0.88
Date: 2026-05-04

Headline: Earnings Strength Takes Center Stage As The Busiest Week Of The Q1 Season Arrives
Sentiment: neutral
Confidence: 0.74
Date: 2026-05-04

Headline: Apple : introduces a new Pride Collection
Sentiment: neutral
Confidence: 0.83

Date: 2026-05-04

Headline: Microsoft Is a Mess. Is the "Magnificent Seven" Stock a Buy in May or Better Avoid
Sentiment: negative
Confidence: 0.95
Date: 2026-05-04

Headline: Apple says users are snapping up Macs faster than ever and created a supply shortage
Sentiment: negative
Confidence: 0.97
Date: 2026-05-04

Headline: Will Bitcoin Ever Be More Valuable Than Apple? You Might Be Surprised at What Predictions Say
Sentiment: positive
Confidence: 0.69
Date: 2026-05-04

Headline: Spirit Airlines Goes Dark: When Geopolitics Lands on the Runway
Sentiment: neutral
Confidence: 0.68
Date: 2026-05-04

Headline: Exploring The Competitive Space: Apple Versus Industry Peers In Technology Hardware
Sentiment: neutral
Confidence: 0.71
Date: 2026-05-04

Headline: Zacks Value Investor Highlights: Apple, Amazon and Starbucks
Sentiment: neutral
Confidence: 0.89
Date: 2026-05-04

Headline: 5 Stocks In The Spotlight Last Week: Wall Street's Most Accurate Analysts Weigh In
Sentiment: negative
Confidence: 0.83
Date: 2026-05-04

Headline: 2 Unpopular Stocks That Deserve Some Love and 1 Facing Challenges
Sentiment: negative
Confidence: 0.93
Date: 2026-05-04

Headline: APPLE INC : Buy rating from Goldman Sachs
Sentiment: neutral
Confidence: 0.93
Date: 2026-05-04

Headline: Apple holds talks with Intel, Samsung for US chipmaking, Bloomberg News reports
Sentiment: negative
Confidence: 0.81
Date: 2026-05-04

Headline: 'Apple Explores Using Intel and Samsung to Build Main Device Chips in the US' - Bloomberg
Sentiment: neutral
Confidence: 0.89
Date: 2026-05-04

Headline: Apple Explores Using Intel and Samsung to Build Main Device Chips in the US
Sentiment: negative
Confidence: 0.55
Date: 2026-05-04

Headline: CNBC Daily Open: Ceasefire at risk of collapse
Sentiment: negative
Confidence: 0.94
Date: 2026-05-04

Headline: Apple Urges Supreme Court To Block Contempt Order In Epic Games Battle-Tim Sweeney
Sentiment: negative
Confidence: 0.92
Date: 2026-05-04

Headline: Is Apple Stock a Buy as iPhone Revenue Soars?
Sentiment: positive
Confidence: 0.88
Date: 2026-05-04

Headline: Intel Joined Elon Musk's Terafab. Will This Catalyst Cement the Stock's Turnaround?
Sentiment: positive
Confidence: 0.94
Date: 2026-05-05

Headline: Apple Talks Put Intel Foundry Ambitions And Investor Hopes In Focus
Sentiment: positive
Confidence: 0.76
Date: 2026-05-05

Headline: PRESS DIGEST-Financial Times - May 6
Sentiment: neutral
Confidence: 0.91
Date: 2026-05-05

Headline: Stock Market Today, May 5: Stocks Rebound as U.S.-Iran Truce Holds
Sentiment: positive
Confidence: 0.95
Date: 2026-05-05

Headline: US Equity Indexes Advance, Crude Oil Drops as Rubio Says Combat Operation Against Iran
Sentiment: negative
Confidence: 0.92
Date: 2026-05-05

Headline: Apple settles lawsuit over late Siri AI features for \$250 million
Sentiment: negative
Confidence: 0.68
Date: 2026-05-05

Headline: Samsung crosses \$1 trillion valuation as AI frenzy drives historic rally, lifting :
Sentiment: positive
Confidence: 0.95
Date: 2026-05-05

Headline: Why Intel Stock Skyrocketed to Record Highs Today
Sentiment: negative
Confidence: 0.83
Date: 2026-05-05

Headline: Stock Market Today: Nasdaq, S&P 500 End At Record Highs As Techs Rise; Intel Shares
Sentiment: positive
Confidence: 0.82
Date: 2026-05-05

Headline: Apple reaches \$250mn settlement over delayed 'AI Siri'
Sentiment: negative
Confidence: 0.83
Date: 2026-05-05

Headline: Nasdaq, S&P 500 Hit Fresh Peaks as Oil Retreats
Sentiment: positive
Confidence: 0.75
Date: 2026-05-05

Headline: S&P 500 and Nasdaq 100 Post Record Highs on Tech Strength
Sentiment: positive
Confidence: 0.93
Date: 2026-05-05

Headline: Skyworks Solutions forecasts higher revenue on strong chip demand
Sentiment: positive
Confidence: 0.91
Date: 2026-05-05

Headline: Beaumont Trims Its JPEF Stake by \$9.0 Million -- Signal or Just Smart Rebalancing?

Sentiment: neutral
Confidence: 0.53
Date: 2026-05-05

Headline: Wednesday's big stock stories: What's likely to move the market in the next trading
Sentiment: positive
Confidence: 0.93
Date: 2026-05-05

Headline: Apple: iPhone And Services Drive Growth, But Margin And Supply Risks Loom
Sentiment: neutral
Confidence: 0.53
Date: 2026-05-05

Headline: Micron stock is surging, hitting a new intraday record high. Why?
Sentiment: positive
Confidence: 0.88
Date: 2026-05-05

Headline: Apple reportedly exploring Intel and Samsung to build chips in the U.S.
Sentiment: neutral
Confidence: 0.76
Date: 2026-05-05

Headline: S&P 500 and Dow Jones Climb; Intel's 14% Surge Steals the Show
Sentiment: neutral
Confidence: 0.88
Date: 2026-05-05

Headline: Intel Has Moved From Laggard to Leader. Where the Chart Is Heading.
Sentiment: positive
Confidence: 0.94
Date: 2026-05-05

Headline: Apple CEO has stark message for Micron stock investors
Sentiment: neutral

Confidence: 0.48
Date: 2026-05-05

Headline: US Equity Indexes Rise, Crude Oil Slides As Iran Ceasefire Stays Intact
Sentiment: negative
Confidence: 0.84
Date: 2026-05-05

Headline: Dow Jones Tech Titan Apple Breaks Out; AI Chipmaker Broadcom In Buy Zone
Sentiment: positive
Confidence: 0.62
Date: 2026-05-05

Headline: Intel Rips As Semiconductor Rally Has Wall Street Seeing Green
Sentiment: positive
Confidence: 0.89
Date: 2026-05-05

Headline: Stock Market Today: Dow Climbs, Nasdaq Leaps As Techs Charge Up; Micron Booms (Live)
Sentiment: positive
Confidence: 0.85
Date: 2026-05-05

Headline: Micron and Intel Shares Push Highs Again: What is Fueling the Rally?
Sentiment: positive
Confidence: 0.88
Date: 2026-05-05

Headline: Trump administration partners with Asian allies to combat chip shortage
Sentiment: positive
Confidence: 0.92
Date: 2026-05-05

Headline: CNBC Markets Now: May 5, 2026
Sentiment: neutral
Confidence: 0.94

Date: 2026-05-05

Headline: Intel Is Rocketing Higher on Reports That It May Partner With Apple. Should You Buy
Sentiment: positive
Confidence: 0.93
Date: 2026-05-05

Headline: 5 Stocks With High ROE to Buy as Markets Battle Intense Volatility
Sentiment: positive
Confidence: 0.92
Date: 2026-05-05

Headline: Veteran fund manager picks best Mag 7 stock for 2026
Sentiment: positive
Confidence: 0.95
Date: 2026-05-05

Headline: Apple reaches \$250mn settlement over delayed 'AI Siri'- FT
Sentiment: negative
Confidence: 0.37
Date: 2026-05-05

Headline: 'Apple reaches \$250mn settlement over delayed AI Siri' - Financial Times
Sentiment: neutral
Confidence: 0.48
Date: 2026-05-05

Headline: Buffett's Legendary Premium Wiped Out - Berkshire Trails S&P 500 By 41 Points Since
Sentiment: negative
Confidence: 0.91
Date: 2026-05-05

Headline: Peer To Peer Network (OTC: PTOP) Announces Deployment of MobiCard(TM) 1.8 to TestF
Sentiment: positive
Confidence: 0.80
Date: 2026-05-05

Headline: Intel stock hits all-time high on Apple chipmaking report
Sentiment: positive
Confidence: 0.51
Date: 2026-05-05

Headline: Top Midday Stories: PayPal Q2 EPS Guide Undershoots Expectations, Plans 20% Workfo
Sentiment: negative
Confidence: 0.85
Date: 2026-05-05

Headline: Apple Hammers The Competition
Sentiment: neutral
Confidence: 0.88
Date: 2026-05-05

Headline: Intel Stock Is Surging. Potential Chip Deal With a Huge Customer Is Providing a Li
Sentiment: positive
Confidence: 0.95
Date: 2026-05-05

Headline: Stock Market Today: Nasdaq Jumps To Record High As Intel Rockets; Oil-
Tracking ETF Tumbles (Live Coverage)
Sentiment: positive
Confidence: 0.58
Date: 2026-05-05

Headline: Jim Cramer Says AI Spending Is No Bubble As Alphabet, Amazon, Apple Surge, While M
Sentiment: positive
Confidence: 0.50
Date: 2026-05-05

Headline: Alphabet closes in on Nvidia's spot as world's biggest company
Sentiment: positive
Confidence: 0.94
Date: 2026-05-05

Headline: Apple, Amazon, TSMC Top List Of Earnings Triple Plays As Beat-And-Raise Stocks Surge
Sentiment: neutral
Confidence: 0.64
Date: 2026-05-05

Headline: Steve Jobs Once Described What Set Humans Apart From Apes And Other 'High Primates' Founder Said This Is What Matters The Most
Sentiment: neutral
Confidence: 0.91
Date: 2026-05-05

Headline: MAGS Is Up 181% Since Launch, but 2026 Reveals Its Hidden Weakness
Sentiment: positive
Confidence: 0.71
Date: 2026-05-05

Headline: Intel Rips 14% Higher: Apple Foundry Reports Crush Bears in Massive Short Squeeze
Sentiment: positive
Confidence: 0.91
Date: 2026-05-05

Headline: Intel Stock Jumps on Potential Apple Chip Deal
Sentiment: positive
Confidence: 0.78
Date: 2026-05-05

Headline: 'Apple to Let Users Choose Rival AI Models Across Its iOS 27 Features' - Bloomberg
Sentiment: neutral
Confidence: 0.88
Date: 2026-05-05

Headline: Sports Technology Industry worth \$68.70 billion by 2030 Amid Strong Industry Growth
Sentiment: neutral
Confidence: 0.63

Date: 2026-05-05

Headline: Alphabet, Intel, Micron Lead Nasdaq Higher as Oil Drops on Paused Mideast Tensions
Sentiment: positive
Confidence: 0.87
Date: 2026-05-05

Headline: PayPal Tumbles 10% Despite Q1 Earnings Beat: Is the Venmo Spin-off Enough to Save the Stock?
Sentiment: negative
Confidence: 0.97
Date: 2026-05-05

Headline: Apple Explores Intel And Samsung As Chip Backup Options
Sentiment: negative
Confidence: 0.66
Date: 2026-05-05

Headline: Bonds In Danger Zone; Market Resilient Despite Iran's Ceasefire Violations
Sentiment: positive
Confidence: 0.75
Date: 2026-05-05

Headline: Baird and Bank of America Raise Apple (AAPL) Price Targets, See Strong Growth Ahead
Sentiment: positive
Confidence: 0.93
Date: 2026-05-05

Headline: Intel Stock Surges on Report of Huge Chip Customer
Sentiment: positive
Confidence: 0.94
Date: 2026-05-05

Headline: Apple's iOS 27 Wallet Upgrade Could Tighten Its iPhone Ecosystem
Sentiment: neutral
Confidence: 0.86

Date: 2026-05-05

Headline: Apple to let users choose rival AI models across iOS 27 features, Bloomberg News r
Sentiment: neutral
Confidence: 0.90
Date: 2026-05-05

Headline: Nasdaq 100 Tops 28,000, Intel Rallies 13%: Stock Market Today
Sentiment: positive
Confidence: 0.94
Date: 2026-05-05

Headline: Apple to let users choose rival AI models across its iOS 27 features - Bloomberg n
Sentiment: positive
Confidence: 0.65
Date: 2026-05-05

Headline: Ford's New EV Plan Already Doomed
Sentiment: neutral
Confidence: 0.78
Date: 2026-05-05

Headline: S&P 500 Rises on Big Tech Bump, Corporate Profits and Oil Price Relief
Sentiment: positive
Confidence: 0.90
Date: 2026-05-05

Headline: UPDATED: 'Apple to Let Users Choose Rival AI Models Across Its iOS 27 Features' - I
Sentiment: neutral
Confidence: 0.90
Date: 2026-05-05

Headline: Apple explores chip production with Intel and Samsung
Sentiment: neutral
Confidence: 0.67
Date: 2026-05-05

Headline: US Equity Futures Gain Pre-Bell as Traders Keep Eye on Potential Re-
Escalation of Middle East Hostilities
Sentiment: positive
Confidence: 0.92
Date: 2026-05-05

Headline: Apple explores Intel, Samsung as U.S. chip suppliers
Sentiment: negative
Confidence: 0.58
Date: 2026-05-05

Headline: Nvidia Stock Is Sitting Out the Latest AI Boom. What Can Get It Moving Again.
Sentiment: negative
Confidence: 0.92
Date: 2026-05-05

Headline: Intel Stock Pops on Report of Huge Chip Customer
Sentiment: positive
Confidence: 0.94
Date: 2026-05-05

Headline: We May Soon Find Out Why Warren Buffett and Berkshire Hathaway Have Almost \$400 Bi
Sentiment: neutral
Confidence: 0.51
Date: 2026-05-05

Headline: The Late Morning Rundown: May 5, 2026
Sentiment: neutral
Confidence: 0.90
Date: 2026-05-05

Headline: Apple Explores Intel, Samsung for U.S. Chipmaking
Sentiment: positive
Confidence: 0.53
Date: 2026-05-05

Headline: Apple explores new chip suppliers beyond TSMC
Sentiment: negative
Confidence: 0.88
Date: 2026-05-05

Headline: Enovix Corporation Appoints Steve Bakos as Senior Vice President of Worldwide Sales
up and Revenue Growth
Sentiment: neutral
Confidence: 0.89
Date: 2026-05-05

Headline: 1 Unstoppable Stock to Buy Before It Joins Nvidia, Alphabet, Microsoft, and Apple :
Sentiment: neutral
Confidence: 0.81
Date: 2026-05-05

Headline: Why Nvidia Stock Is 'Undervalued' Amid Soaring AI Demand
Sentiment: negative
Confidence: 0.94
Date: 2026-05-05

Headline: Intel soars 14% on report of Apple chip talks, hits new all-time high
Sentiment: positive
Confidence: 0.92
Date: 2026-05-05

Headline: Intel Shares Rise as Apple Explores Expanding Chip Supply Chain
Sentiment: positive
Confidence: 0.82
Date: 2026-05-05

Headline: Apple explores alternatives to TSMC to secure supply chains
Sentiment: positive
Confidence: 0.81
Date: 2026-05-05

Headline: These Stocks Are Today's Movers: Palantir, Intel, AMD, ON Semi, Pinterest, Duolingo
Sentiment: negative
Confidence: 0.73
Date: 2026-05-05

Headline: Here's What BofA Thinks About NVIDIA Corporation (NVDA)
Sentiment: neutral
Confidence: 0.76
Date: 2026-05-05

Headline: U.S.-Iran Ceasefire Under Pressure; AMD Earnings in Focus: Dow Jones, S&P, Nasdaq,
Sentiment: positive
Confidence: 0.92
Date: 2026-05-05

Headline: APPLE INC : UBS is Neutral
Sentiment: neutral
Confidence: 0.79
Date: 2026-05-05

Headline: TSMC Stock Holds Firm As \$56 Billion AI Expansion Defies Apple Supply Chain Rumors
Sentiment: negative
Confidence: 0.90
Date: 2026-05-05

Headline: Earnings Triple Plays Piling Up In Q1 2026
Sentiment: positive
Confidence: 0.90
Date: 2026-05-05

Headline: Apple : Manufacturing Academy accelerates AI use in U.S. supply chains
Sentiment: positive
Confidence: 0.77
Date: 2026-05-05

Headline: Apple Crushed Q2 Earnings, but Tim Cook Just Issued a Warning About This Looming Threat
Sentiment: negative
Confidence: 0.82
Date: 2026-05-05

Headline: Ford's secret EV unit emerges from shadows, still bullish on new pickup amid market uncertainty
Sentiment: positive
Confidence: 0.58
Date: 2026-05-05

Headline: What's Going On With Intel Stock Tuesday?
Sentiment: neutral
Confidence: 0.81
Date: 2026-05-05

Headline: Apple: A Once-In-A-Generation Buy Signal Just Flashed
Sentiment: positive
Confidence: 0.95
Date: 2026-05-05

Headline: Apple's chip talks with Intel and Samsung signal a structural shift in how Big Tech competes
Sentiment: neutral
Confidence: 0.62
Date: 2026-05-05

Headline: Stocks making the biggest moves premarket: Palantir Technologies, PayPal, Intel, Amazon, and
Busch InBev & more
Sentiment: neutral
Confidence: 0.88
Date: 2026-05-05

Headline: Stock futures are higher as Wall Street eyes latest developments in Middle East
Sentiment: neutral
Confidence: 0.41
Date: 2026-05-05

Headline: iPhone 17 Rules, Becomes World's Best-Selling Smartphone
Sentiment: neutral
Confidence: 0.72
Date: 2026-05-05

Headline: Competitor Analysis: Evaluating Apple And Competitors In Technology Hardware, Storage
Sentiment: neutral
Confidence: 0.90
Date: 2026-05-05

Headline: What's Going On With Apple Stock Tuesday?
Sentiment: neutral
Confidence: 0.54
Date: 2026-05-05

Headline: Bank of America resets Apple stock price target after earnings
Sentiment: positive
Confidence: 0.93
Date: 2026-05-05

Headline: Roundhill Files for 'Magnificent Seven Plus' ETF With AI, Space Twist
Sentiment: neutral
Confidence: 0.82
Date: 2026-05-05

Headline: Apple Eyes US Chip Production With Intel, Samsung To Reduce TSMC Reliance: Report
Sentiment: positive
Confidence: 0.90
Date: 2026-05-05

Headline: S&P 500 Hits New High As Oil Prices Fall: Investor Sentiment Improves, Fear Index I
Sentiment: positive
Confidence: 0.86
Date: 2026-05-05

Headline: Samsung Hits \$1 Trillion As KOSPI Tops 7,300 And Korean ETFs Post Record Gains: Wh
Sentiment: positive
Confidence: 0.95
Date: 2026-05-05

Headline: CNBC Daily Open: Kospi bull run rolls on
Sentiment: positive
Confidence: 0.93
Date: 2026-05-05

Headline: Apple is facing delivery problems
Sentiment: positive
Confidence: 0.95
Date: 2026-05-05

Headline: Tracking Ken Fisher's Fisher Asset Management Portfolio - Q1 2026 Update
Sentiment: neutral
Confidence: 0.93
Date: 2026-05-06

Headline: The smartphone era is ending the way it started: with Apple
Sentiment: neutral
Confidence: 0.89
Date: 2026-05-06

Headline: Apple Stock Hit a Record High. Why June Could Change Everything.
Sentiment: positive
Confidence: 0.88
Date: 2026-05-06

Headline: Arm stock slides as chip supply worries overshadow upbeat AI-
fueled revenue forecast
Sentiment: negative
Confidence: 0.97
Date: 2026-05-06

Headline: Arm forecasts higher-than-expected revenue on surging AI data center demand
Sentiment: negative
Confidence: 0.96
Date: 2026-05-06

Headline: Dear Apple Stock Fans, Mark Your Calendars for May 11
Sentiment: positive
Confidence: 0.64
Date: 2026-05-06

Headline: Samsung market cap crosses \$1 trillion amid memory chip rally
Sentiment: negative
Confidence: 0.59
Date: 2026-05-06

Headline: What Apple (AAPL)'s Massive Buyback And Chip Shift Means For Shareholders
Sentiment: positive
Confidence: 0.87
Date: 2026-05-06

Headline: If you're an iPhone user, you could get \$95 from this Apple settlement
Sentiment: negative
Confidence: 0.83
Date: 2026-05-06

Headline: The Smartest Stocks to Buy With \$1,000 in May
Sentiment: neutral
Confidence: 0.83
Date: 2026-05-06

Headline: March Quarter Results Bolster Confidence in Apple Inc (AAPL) Stock
Sentiment: positive
Confidence: 0.95
Date: 2026-05-06

Headline: Apple Stock Tracking a Record High. Why June Could Change Everything.

Sentiment: positive
Confidence: 0.88
Date: 2026-05-06

Headline: Semi Stocks Are Flying. Why Skyworks Solutions Is Getting Crushed.
Sentiment: negative
Confidence: 0.97
Date: 2026-05-06

Headline: Forget Tariffs: An Apple-Intel Deal Could Be the Biggest Manufacturing Story of the Year
Sentiment: neutral
Confidence: 0.86
Date: 2026-05-06

Headline: Stay informed about the most active stocks in the S&P500 index on Wednesday's session
Sentiment: neutral
Confidence: 0.88
Date: 2026-05-06

Headline: Intel Stock Just Hit New Record Highs. This Time, Apple Gets All the Credit.
Sentiment: positive
Confidence: 0.70
Date: 2026-05-06

Headline: Apple Opens AI Platform To Rivals Across 2 Billion Devices
Sentiment: neutral
Confidence: 0.61
Date: 2026-05-06

Headline: iPhone owners could get up to \$95 from a new Apple settlement
Sentiment: positive
Confidence: 0.87
Date: 2026-05-06

Headline: Analyzing Apple In Comparison To Competitors In Technology Hardware, Storage & Services
Sentiment: neutral

Confidence: 0.71
Date: 2026-05-06

Headline: Apple class action settlement could pay you money. See who's eligible
Sentiment: neutral
Confidence: 0.88
Date: 2026-05-06

Headline: Apple to pay 250 million dollars to settle deceptive AI advertising lawsuit
Sentiment: positive
Confidence: 0.86
Date: 2026-05-06

Headline: The space ETFs prepping for liftoff ahead of SpaceX IPO
Sentiment: positive
Confidence: 0.63
Date: 2026-05-06

Headline: Apple Explores New Chip Suppliers As Investors Weigh Supply Chain Risks
Sentiment: positive
Confidence: 0.92
Date: 2026-05-06

Headline: Apple to pay \$250M to settle lawsuit over Siri's delayed AI features
Sentiment: negative
Confidence: 0.78
Date: 2026-05-06

Headline: Apple to Let Users Choose Rival AI Models
Sentiment: neutral
Confidence: 0.74
Date: 2026-05-06

Headline: Samsung Hits \$1 Trillion As AI Chip Demand Fuels 14% Surge
Sentiment: positive
Confidence: 0.94

Date: 2026-05-06

Headline: Apple Just Discontinued Its Cheapest Mac Mini, but Wedbush Says It's Good News for
Sentiment: negative
Confidence: 0.97
Date: 2026-05-06

Headline: US Supreme Court declines to pause order holding Apple in contempt in Epic Games l
Sentiment: negative
Confidence: 0.83
Date: 2026-05-06

Headline: TSMC Stock Rallies: \$56 Billion AI Expansion Crushes Apple Supply Chain Fears
Sentiment: positive
Confidence: 0.72
Date: 2026-05-06

Headline: While Big Tech Spends \$725 Billion Building AI, Apple Is Letting Them. And That May
Sentiment: neutral
Confidence: 0.89
Date: 2026-05-06

Headline: AI boom pushes Samsung to \$1T
Sentiment: positive
Confidence: 0.94
Date: 2026-05-06

Headline: 10 Information Technology Stocks With Whale Alerts In Today's Session
Sentiment: neutral
Confidence: 0.78
Date: 2026-05-06

Headline: Ford Has a Secret Team Working on a \$30,000 Electric Truck to Beat China. It's a 'P
Sentiment: neutral
Confidence: 0.85
Date: 2026-05-06

Headline: Apple settles lawsuit tied to delayed Siri AI features
Sentiment: negative
Confidence: 0.84
Date: 2026-05-06

Headline: Arm Is Shifting Its Business Model. Earnings Will Be a Test for the Stock.
Sentiment: negative
Confidence: 0.77
Date: 2026-05-06

Headline: US device promotions: the subsidy wrapper hiding the real price-service revenue
Sentiment: negative
Confidence: 0.97
Date: 2026-05-06

Headline: Apple faces supply strain from strong Mac demand
Sentiment: negative
Confidence: 0.82
Date: 2026-05-06

Headline: Apple settles \$250 million Siri false advertising lawsuit
Sentiment: neutral
Confidence: 0.76
Date: 2026-05-06

Headline: India's top factory hub hopeful, wary as film star Vijay set to take reins
Sentiment: neutral
Confidence: 0.69
Date: 2026-05-06

Headline: Is Semi Mania Leading To A 2000-Style Crash? US-Iran Proposal Lifts Market, Pressures
Sentiment: negative
Confidence: 0.73
Date: 2026-05-06

Headline: Apple Settles Siri AI Delay Lawsuit for \$250 Million
Sentiment: neutral
Confidence: 0.50
Date: 2026-05-06

Headline: Can the Rest of the Mag 7 Reach the Google-Level Greatness Shown Last Quarter?
Sentiment: neutral
Confidence: 0.51
Date: 2026-05-06

Headline: Apple, Nvidia, Broadcom And This Dow Retail Behemoth Fire Up Breakout Watch
Sentiment: positive
Confidence: 0.55
Date: 2026-05-06

Headline: Apple: This Is More Than A Quarterly Peak
Sentiment: positive
Confidence: 0.95
Date: 2026-05-06

Headline: iPhone 17 Rules, Becomes World's Best-Selling Smartphone
Sentiment: positive
Confidence: 0.93
Date: 2026-05-06

Headline: Alphabet's \$346 Billion Investment Has Fueled Its 1,000% Rally Over the Last 10 Years
Sentiment: positive
Confidence: 0.95
Date: 2026-05-06

Headline: Apple Explores Intel And Samsung As Chip Constraints Hit Growth
Sentiment: negative
Confidence: 0.64
Date: 2026-05-06

Headline: Some iPhone owners could get up to \$95 payment after Apple agrees to settle case f
Sentiment: negative
Confidence: 0.85
Date: 2026-05-06

Headline: Despite higher energy prices, there's an opportunity in travel stocks, says Janus'
Sentiment: neutral
Confidence: 0.79
Date: 2026-05-06

Headline: U.S. Supreme Court declines to pause order holding Apple in contempt in antitrust l
Sentiment: negative
Confidence: 0.70
Date: 2026-05-06

Headline: U.S. Supreme Court Declines To Pause Order Holding Apple In Contempt In Antitrust l
Sentiment: negative
Confidence: 0.80
Date: 2026-05-06

Headline: What's Going On With Intel Stock Wednesday?
Sentiment: positive
Confidence: 0.80
Date: 2026-05-06

Headline: Zacks.com featured highlights include Ross Stores, TE Connectivity, Broadcom, Corn
Sentiment: positive
Confidence: 0.89
Date: 2026-05-06

Headline: Zacks Investment Ideas feature highlights: Micron Technology and Intel
Sentiment: positive
Confidence: 0.90
Date: 2026-05-06

Headline: FuboTV Loses 500,000 Subscribers In Q2

Sentiment: negative
Confidence: 0.89
Date: 2026-05-06

Headline: Berkshire Hathaway: Don't Worry Critics, Greg Abel Has The Right Kind Of Investment
Sentiment: neutral
Confidence: 0.78
Date: 2026-05-06

Headline: Tariff Refunds Are Coming, And They Could Keep Interest Rates Higher For Longer
Sentiment: neutral
Confidence: 0.83
Date: 2026-05-06

Headline: Peace hopes, Disney earnings, dismantling Spirit Airlines and more in Morning Square
Sentiment: neutral
Confidence: 0.91
Date: 2026-05-06

Headline: Apple's R&D investments top 10% of sales as AI race creates 'sense of urgency'
Sentiment: positive
Confidence: 0.95
Date: 2026-05-06

Headline: Record Highs, Punished Results, and One Very Consequential Apple Rumor
Sentiment: negative
Confidence: 0.93
Date: 2026-05-06

Headline: Samsung Hits \$1 Trillion Valuation, Joining TSMC in Elite Club
Sentiment: positive
Confidence: 0.62
Date: 2026-05-06

Headline: Nvidia to invest up to \$3.2 billion in Corning as part of massive optical fiber deal
Sentiment: neutral

Confidence: 0.69
Date: 2026-05-06

Headline: Wall Street Breakfast Podcast: Soda & Chips: McDonald's And AMD In Focus
Sentiment: positive
Confidence: 0.86
Date: 2026-05-06

Headline: Arm Reports Earnings as It Shifts Its Business Model
Sentiment: negative
Confidence: 0.68
Date: 2026-05-06

Headline: Qualcomm Stock's Momentum Score Surges Amid AI Optimism And Data Center Growth Plan
Sentiment: positive
Confidence: 0.95
Date: 2026-05-06

Headline: CNBC Daily Open: Iran reviews 'one-page peace plan'
Sentiment: positive
Confidence: 0.57
Date: 2026-05-06

Headline: The Only Fully Integrated U.S. Rare Earth Producer Has a Deal With Apple and the P
Sentiment: positive
Confidence: 0.68
Date: 2026-05-06

Headline: Anthropic Is Making Its Claude Chatbot More Appealing to Consumers
Sentiment: neutral
Confidence: 0.66
Date: 2026-05-06

Headline: The 2026 Chip War Is Already Over and Apple Dominated
Sentiment: neutral
Confidence: 0.88

Date: 2026-05-07

Headline: Apple Legal Challenges Put App Store Power And AI Claims In Focus

Sentiment: negative

Confidence: 0.76

Date: 2026-05-07

Headline: Why Invesco NASDAQ 100 ETF's Lower Fee Quietly Outperforms Invesco QQQ Trust for Buy-and-Hold Investors

Sentiment: neutral

Confidence: 0.90

Date: 2026-05-07

Headline: Tech Stocks Face a New AI Hurdle: A Widespread Components Shortage

Sentiment: negative

Confidence: 0.96

Date: 2026-05-07

Headline: iShares Core S&P Total U.S. Stock Market ETF's 0.03% Fee and BlackRock Scale Face Off

Sentiment: neutral

Confidence: 0.87

Date: 2026-05-07

Headline: Trump meets with Brazil's Lula to talk trade, extends EU tariff deadline

Sentiment: neutral

Confidence: 0.84

Date: 2026-05-07

Headline: Top Stock Reports for Apple, AMD & Chevron

Sentiment: positive

Confidence: 0.94

Date: 2026-05-07

Headline: Apple expands renewable energy, recycling initiatives in India

Sentiment: positive

Confidence: 0.85

Date: 2026-05-07

Headline: They Asked Value Investors Which Stock They'd Put Their Entire Net Worth For 10 Years
Sentiment: neutral
Confidence: 0.85
Date: 2026-05-07

Headline: Jim Cramer sits down with the CEOs of Nvidia and Corning to talk partnership
Sentiment: neutral
Confidence: 0.73
Date: 2026-05-07

Headline: A \$100 Billion Reason to Buy Apple Stock Here
Sentiment: neutral
Confidence: 0.62
Date: 2026-05-07

Headline: Stocks take a breather after reaching record highs
Sentiment: neutral
Confidence: 0.80
Date: 2026-05-07

Headline: Friday's big stock stories: What's likely to move the market in the next trading session
Sentiment: negative
Confidence: 0.97
Date: 2026-05-07

Headline: AI Is Making Smartphones More Expensive-But Not For the Reason You Think
Sentiment: negative
Confidence: 0.97
Date: 2026-05-07

Headline: Some iPhone Owners Could Get Payouts Under a \$250M Settlement-Here's What You Need to Know
Sentiment: neutral
Confidence: 0.62
Date: 2026-05-07

Headline: EWT, EWY, and SOXX: The Only 3 ETFs You Need for Semiconductor Dominance
Sentiment: neutral
Confidence: 0.92
Date: 2026-05-07

Headline: Apple's Camera-Equipped AirPods Reach Late Testing in AI Device Push
Sentiment: positive
Confidence: 0.61
Date: 2026-05-07

Headline: What's going on in today's session: S&P500 most active stocks
Sentiment: neutral
Confidence: 0.92
Date: 2026-05-07

Headline: New Logitech Rugged Combo 4c Keyboard Cases Deliver Drop-Tested Durability and New C Connectivity to Maximize Classroom Time
Sentiment: positive
Confidence: 0.55
Date: 2026-05-07

Headline: Apple at \$284: Buy, Sell or Hold?
Sentiment: positive
Confidence: 0.71
Date: 2026-05-07

Headline: What Long-Term Investors Should Know Before Buying This Broad Market ETF
Sentiment: neutral
Confidence: 0.93
Date: 2026-05-07

Headline: Apple helps 'Magnificent 7' shake off its bear market scare
Sentiment: neutral
Confidence: 0.72
Date: 2026-05-07

Headline: Apple warns Canadian bill could force it to weaken device encryption
Sentiment: negative
Confidence: 0.77
Date: 2026-05-07

Headline: Apple Stock Just Broke to New Record Highs with \$300 Just Around the Corner
Sentiment: positive
Confidence: 0.95
Date: 2026-05-07

Headline: True North Boosts Flexible Bond Position by \$6 Million Amid Rate Volatility
Sentiment: positive
Confidence: 0.94
Date: 2026-05-07

Headline: Rave files antitrust lawsuit against Apple over removal of video-sharing app
Sentiment: negative
Confidence: 0.90
Date: 2026-05-07

Headline: 'Apple's Camera-Equipped AirPods Reach Late Testing in AI Device Push' - Bloomberg
Sentiment: neutral
Confidence: 0.86
Date: 2026-05-07

Headline: Rave Sues Apple in Five Countries Over App Store Removal
Sentiment: negative
Confidence: 0.93
Date: 2026-05-07

Headline: It Takes Meta Almost 9 Minutes to Make \$1 Million in Profits. Google Does It in Un
Sentiment: neutral
Confidence: 0.90
Date: 2026-05-07

Headline: Apple Stock Notches Record High Ahead Of WWDC News
Sentiment: positive
Confidence: 0.95
Date: 2026-05-07

Headline: Google Launches \$100 Fitbit Air Without a Screen to Rival Whoop
Sentiment: neutral
Confidence: 0.76
Date: 2026-05-07

Headline: Is Intuitive Machines Stock a Buy After Once Capital Initiated a New Position?
Sentiment: neutral
Confidence: 0.73
Date: 2026-05-07

Headline: This Sneaky AI Play Nearly Doubles Sales. The Stock Is Rising 32%.
Sentiment: positive
Confidence: 0.95
Date: 2026-05-07

Headline: Stock Market Today: Dow Rises After Jobless Claims; AI Stock Dives On Earnings (Li
Sentiment: negative
Confidence: 0.93
Date: 2026-05-07

Headline: How filmmakers are redefining the art form with MAMI Select: Filmed on iPhone
Sentiment: neutral
Confidence: 0.87
Date: 2026-05-07

Headline: Apple's AI Restraint Looks Smart as Microsoft Spends \$190B on AI Despite User Push
Sentiment: neutral
Confidence: 0.53
Date: 2026-05-07

Headline: Navigating the AI-led stock market
Sentiment: neutral
Confidence: 0.94
Date: 2026-05-07

Headline: 'Trump administration plans to invite CEOs from Nvidia, Apple, Exxon on China trip
Sentiment: neutral
Confidence: 0.94
Date: 2026-05-07

Headline: The MacBook Neo Could Be the Best Reason to Buy Apple Stock
Sentiment: neutral
Confidence: 0.64
Date: 2026-05-07

Headline: Stocks Hit Record Highs Ahead Of Jobs Report, Trump's China Visit
Sentiment: positive
Confidence: 0.52
Date: 2026-05-07

Headline: Nvidia, Apple Drive A Mag 7 Coup That Just Seized 55% Of S&P 500 Profits
Sentiment: neutral
Confidence: 0.89
Date: 2026-05-07

Headline: InMobi Acquires MobileAction, a Leading AI Platform for iOS growth
Sentiment: positive
Confidence: 0.85
Date: 2026-05-07

Headline: State Street Calls It "The first \$1 trillion ETF will land this year"
Sentiment: neutral
Confidence: 0.88
Date: 2026-05-07

Headline: 'Rave Inc. Files Antitrust Suit Against Apple In U.S. Federal Court Alleging Apple
Sentiment: negative
Confidence: 0.93
Date: 2026-05-07

Headline: Rave files antitrust lawsuit against Apple over removal of video-
sharing app from App Store
Sentiment: negative
Confidence: 0.95
Date: 2026-05-07

Headline: Arm beats expectations but shares fall on supply concerns for AI chip
Sentiment: negative
Confidence: 0.95
Date: 2026-05-07

Headline: Iran Talks, Fed Rates, Jobs Data, Inflation. This Market Rally Faces Big Tests Soon
Sentiment: neutral
Confidence: 0.70
Date: 2026-05-07

Headline: Trade Strategy For SPY, AAPL, MSFT, NVDA, GOOGL, META, And TSLA
Sentiment: neutral
Confidence: 0.95
Date: 2026-05-07

Headline: These Stocks Are Today's Movers: Arm, AMD, Apple, Fortinet, McDonald's, Whirlpool,
Sentiment: negative
Confidence: 0.96
Date: 2026-05-07

Headline: AI Chip Market Research and Global Forecast Report 2025-2030: Lucrative Opportunities
Driven by AI Chip Market Expansion, Cloud Infrastructure Expansion and Rising Adoption of ASIC Tech
Sentiment: positive
Confidence: 0.84
Date: 2026-05-07

Headline: An Ode To Restraint: Lessons From The Tim Cook Legacy
Sentiment: neutral
Confidence: 0.87
Date: 2026-05-07

Headline: Apple : AI meets accessibility in this year's Swift Student Challenge
Sentiment: neutral
Confidence: 0.91
Date: 2026-05-07

Headline: Amazon, Google, Microsoft In Focus As EU Reportedly Mulls Restricting US Cloud Pro
Sentiment: negative
Confidence: 0.86
Date: 2026-05-07

Headline: Intel Stock Bubble Warning - More Extreme Than Cisco At Dot-Com Peak
Sentiment: negative
Confidence: 0.83
Date: 2026-05-07

Headline: Family office deal-making rebounds in April with healthcare bets
Sentiment: negative
Confidence: 0.88
Date: 2026-05-07

Headline: Apple Wants To Fix The One Thing Users Hate Most About Foldables
Sentiment: neutral
Confidence: 0.77
Date: 2026-05-07

Headline: Arm shares slides as concerns over AI chip supply overshadow upbeat forecast
Sentiment: negative
Confidence: 0.97
Date: 2026-05-07

Headline: This "Magnificent Seven" Stock Is the Cheapest of Them All. Is It a Buy Right Now?
Sentiment: neutral
Confidence: 0.89
Date: 2026-05-07

Headline: Coke Keeps Beating Pepsi. Warren Buffett's Bet Has Paid Off.
Sentiment: positive
Confidence: 0.86
Date: 2026-05-07

Headline: AEF: Profit From The AI Spending Binge With This Emerging Market Fund
Sentiment: neutral
Confidence: 0.84
Date: 2026-05-07

The transformation stage focused on converting raw financial news data into a clean and structured dataset suitable for sentiment analysis and exploratory data analysis (EDA). The raw data was extracted from the Finnhub API and initially contained nested JSON objects with unnecessary metadata, inconsistent formatting, and timestamp values that were not human-readable. Several transformation steps were performed to improve data quality, consistency, and usability. First, relevant attributes were selected from the raw API response, including the stock ticker symbol, article headline, summary, source, and publication date. This reduced unnecessary information and ensured the final dataset only contained variables relevant to the project objectives.

Next, the headline and summary fields were combined into a single text field. This was done because sentiment analysis models generally perform better with more contextual information rather than short headlines alone. Combining both fields allowed the sentiment model to capture additional financial context from the article summaries.

Text-cleaning transformations were then applied to standardize the article content. Special characters such as non-breaking spaces and HTML escape characters were removed or replaced to improve readability and prevent encoding issues during analysis. Extra whitespace was also removed using string normalization techniques. Additionally, UTF-8 encoding and decoding were applied to handle malformed characters and ensure consistent text formatting across all articles.

The Unix timestamp values provided by Finnhub were transformed into human-readable dates using Python's datetime module. This made the dataset easier to interpret and enabled later temporal analysis of sentiment trends over time.

After cleaning and restructuring the data, the transformed text was processed using the FinBERT sentiment analysis model from Hugging Face Transformers. FinBERT is specifically

trained on financial text, making it more appropriate for financial news sentiment classification than general-purpose sentiment models. Each article received a sentiment label (positive, negative, or neutral) along with a confidence score representing the model's certainty.

Finally, all transformed and enriched records were stored in a list and converted into a Pandas DataFrame. This final structured dataset served as the project's final data product and was later used for descriptive and bias-related exploratory data analysis.

The transformation decisions were guided by the project goal of analyzing relationships between financial news sentiment and stock-related discussion trends. Ensuring clean, standardized, and context-rich text data was important for improving the reliability of the sentiment analysis results and supporting accurate downstream analysis.

Analysis & Evaluation

Descriptive EDA Statistics

```
print("\n=====")
print("DESCRIPTIVE EDA STATISTICS")
print("=====")

# Basic info
print("\nDataset shape:", df.shape)
print("\nMissing values:\n", df.isnull().sum())

# Ticker distribution
print("\nTicker distribution:\n", df["ticker"].value_counts())

# Sentiment distribution
print("\nSentiment distribution:\n", df["sentiment"].value_counts())

# Source distribution (bias indicator)
print("\nSource distribution:\n", df["source"].value_counts())

# Confidence summary
# 50% is the median value
print("\nConfidence stats:\n", df["sen_confidence"].describe())

# Sentiment distribution
print("\nSentiment distribution:\n", df["sentiment"].value_counts())

# Text columns info
```

```

# Headline
# Usually a short, concise news title
# Typically 5-20 words
# Designed to summarize the main event quickly

# Summary
# A longer paragraph-style description of the article
# Typically 20-100+ words
# Provides key financial/business events and more context

# Text
# Headline + summary concatenated (combined)
# At least as long as the headline
# Usually longer than just the summary
# Used as input for the FinBERT sentiment model

text_columns = ["headline", "summary", "text"]

summary_stats = []

for col in text_columns:

    word_counts = df[col].fillna("").apply(
        lambda x: len(str(x).split())
    )

    summary_stats.append({
        "column": col,
        "min": word_counts.min(),
        "max": word_counts.max(),
        "mean": round(word_counts.mean(), 2),
        "median": word_counts.median(),
        "missing": df[col].isnull().sum()
    })

stats_df = pd.DataFrame(summary_stats)
print("\nStats for text columns:")
print(stats_df)

# Date column info
df["date"] = pd.to_datetime(df["date"])

```

```

earliest_date = df["date"].min()
latest_date = df["date"].max()

print("\nDate stats:")
print("min ", earliest_date)
print("max ", latest_date)
print("timespan ", latest_date - earliest_date)
print("median ", df["date"].median())
print("mean ", df["date"].mean())

print("\nDate distribution:")
print(df["date"].value_counts())

# Sentiment over time
print("\nSentiment by date:")
print(df.groupby("date")["sentiment"].value_counts())

# Average confidence per sentiment
print("\nAvg confidence by sentiment:")
print(df.groupby("sentiment")["sen_confidence"].mean())

```

```

=====
DESCRIPTIVE EDA STATISTICS
=====

```

Dataset shape: (434, 8)

Missing values:

ticker	0
headline	0
summary	0
text	0
source	0
date	0
sentiment	0
sen_confidence	0

dtype: int64

Ticker distribution:

ticker	
AAPL	434

Name: count, dtype: int64

Sentiment distribution:

sentiment

positive 185

neutral 154

negative 95

Name: count, dtype: int64

Source distribution:

source

Yahoo 270

Benzinga 70

CNBC 36

Finnhub 28

SeekingAlpha 20

ChartMill 10

Name: count, dtype: int64

Confidence stats:

count 434.000000

mean 0.825678

std 0.134088

min 0.374870

25% 0.758610

50% 0.879218

75% 0.930845

max 0.975517

Name: sen_confidence, dtype: float64

Sentiment distribution:

sentiment

positive 185

neutral 154

negative 95

Name: count, dtype: int64

Stats for text columns:

	column	min	max	mean	median	missing
0	headline	4	27	11.83	11.0	0
1	summary	0	285	32.56	23.0	0
2	text	8	293	44.39	35.0	0

Date stats:
min 2026-05-01 00:00:00
max 2026-05-07 00:00:00
timespan 6 days 00:00:00
median 2026-05-05 00:00:00
mean 2026-05-03 22:20:27.649769

Date distribution:
date
2026-05-01 149
2026-05-05 101
2026-05-06 68
2026-05-07 60
2026-05-04 56
Name: count, dtype: int64

Sentiment by date:
date sentiment
2026-05-01 positive 82
neutral 42
negative 25
2026-05-04 neutral 27
negative 17
positive 12
2026-05-05 positive 52
neutral 32
negative 17
2026-05-06 positive 26
neutral 22
negative 20
2026-05-07 neutral 31
negative 16
positive 13
Name: count, dtype: int64

Avg confidence by sentiment:
sentiment
negative 0.828698
neutral 0.805133
positive 0.841229
Name: sen_confidence, dtype: float64

Descriptive EDA Visualizations

```

# Confidence distribution histogram

plt.figure(figsize=(8,5))

sns.histplot(
    df["sen_confidence"],
    bins=20,
    kde=True
)

plt.title("Distribution of FinBERT Confidence Scores")
plt.xlabel("Confidence")
plt.ylabel("Frequency")
plt.show()

# Ticker distribution bar graph

plt.figure(figsize=(6,4))

sns.countplot(
    data=df,
    x="ticker",
    order=df["ticker"].value_counts().index
)

plt.title("Ticker Distribution")
plt.xlabel("Ticker")
plt.ylabel("Count")
plt.show()

# Sentiment distribution bar graph

plt.figure(figsize=(6,4))

sns.countplot(
    data=df,
    x="sentiment",
    order=df["sentiment"].value_counts().index
)

plt.title("Sentiment Distribution")
plt.xlabel("Sentiment")

```

```

plt.ylabel("Count")
plt.show()

# Source distribution bar graph

plt.figure(figsize=(10,6))

source_counts = df["source"].value_counts().head(10) # up to 10 sources

sns.barplot(
    x=source_counts.values,
    y=source_counts.index
)

plt.title("Top News Sources")
plt.xlabel("Article Count")
plt.ylabel("Source")

plt.show()

# Word count histograms for text columns

df["headline_word_count"] = df["headline"].fillna("").apply(
    lambda x: len(str(x).split())
)

df["summary_word_count"] = df["summary"].fillna("").apply(
    lambda x: len(str(x).split())
)

df["text_word_count"] = df["text"].fillna("").apply(
    lambda x: len(str(x).split())
)

plt.figure(figsize=(8,5))

sns.histplot(
    df["headline_word_count"],
    bins=15,
    kde=True
)

```

```

plt.title("Headline Word Count Distribution")
plt.xlabel("Words")
plt.ylabel("Frequency")
plt.show()

plt.figure(figsize=(8,5))

sns.histplot(
    df["summary_word_count"],
    bins=20,
    kde=True
)

plt.title("Summary Word Count Distribution")
plt.xlabel("Words")
plt.ylabel("Frequency")

plt.show()

plt.figure(figsize=(8,5))

sns.histplot(
    df["text_word_count"],
    bins=20,
    kde=True
)

plt.title("Combined Text Word Count Distribution")
plt.xlabel("Words")
plt.ylabel("Frequency")
plt.show()

# Date frequencies bar graph

plt.figure(figsize=(10,5))

date_counts = df["date"].value_counts().sort_index()

sns.barplot(
    x=date_counts.index.astype(str),
    y=date_counts.values
)

```

```

plt.xticks(rotation=45)
plt.title("Articles Published Per Day")
plt.xlabel("Date")
plt.ylabel("Article Count")

plt.show()

# Confidence by sentiment box and whisker plots

plt.figure(figsize=(8,5))

sns.boxplot(
    data=df,
    x="sentiment",
    y="sen_confidence"
)

plt.title("Confidence by Sentiment")
plt.xlabel("Sentiment")
plt.ylabel("Confidence Score")
plt.show()

# Sentiment by date

df["date"] = pd.to_datetime(df["date"])

sentiment_time = pd.crosstab(
    df["date"],
    df["sentiment"]
)

sentiment_time.index = sentiment_time.index.strftime("%Y-%m-%d")

sentiment_time.plot(
    kind="bar",
    stacked=False,
    figsize=(10,6)
)

plt.title("Sentiment Distribution Over Time")
plt.xlabel("Date")
plt.ylabel("Article Count")

```

```

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Sentiment by news source bar graph

top_sources = df["source"].value_counts().head(5).index

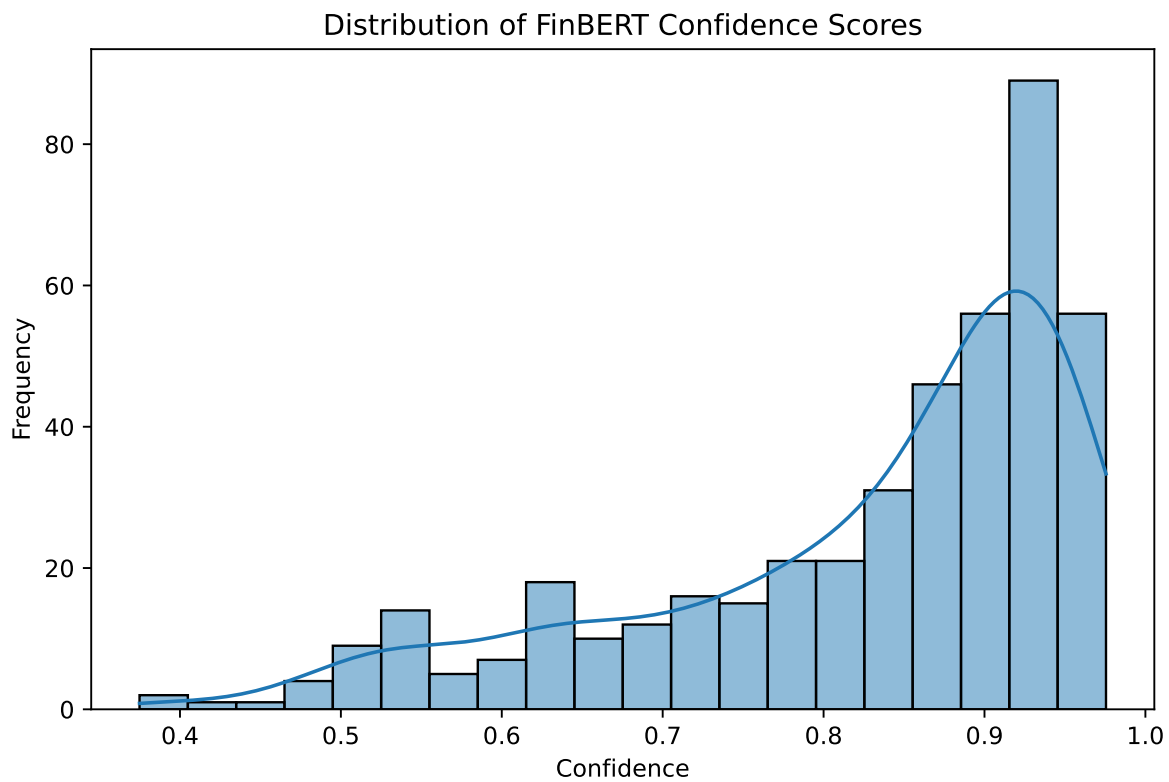
filtered_df = df[df["source"].isin(top_sources)]

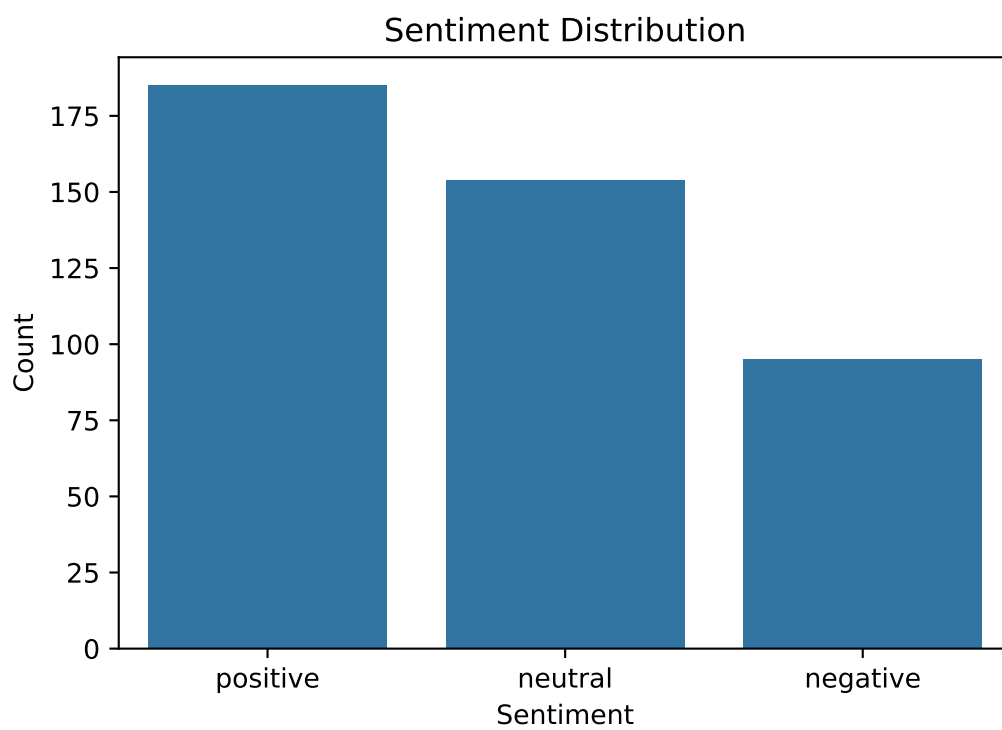
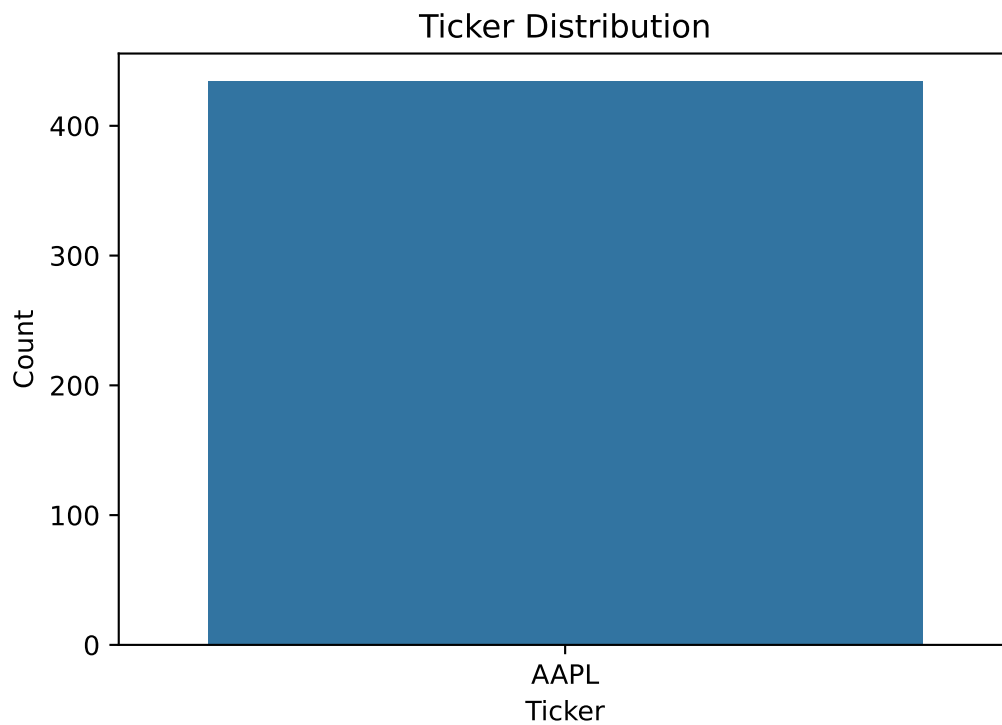
source_sentiment = pd.crosstab(
    filtered_df["source"],
    filtered_df["sentiment"]
)

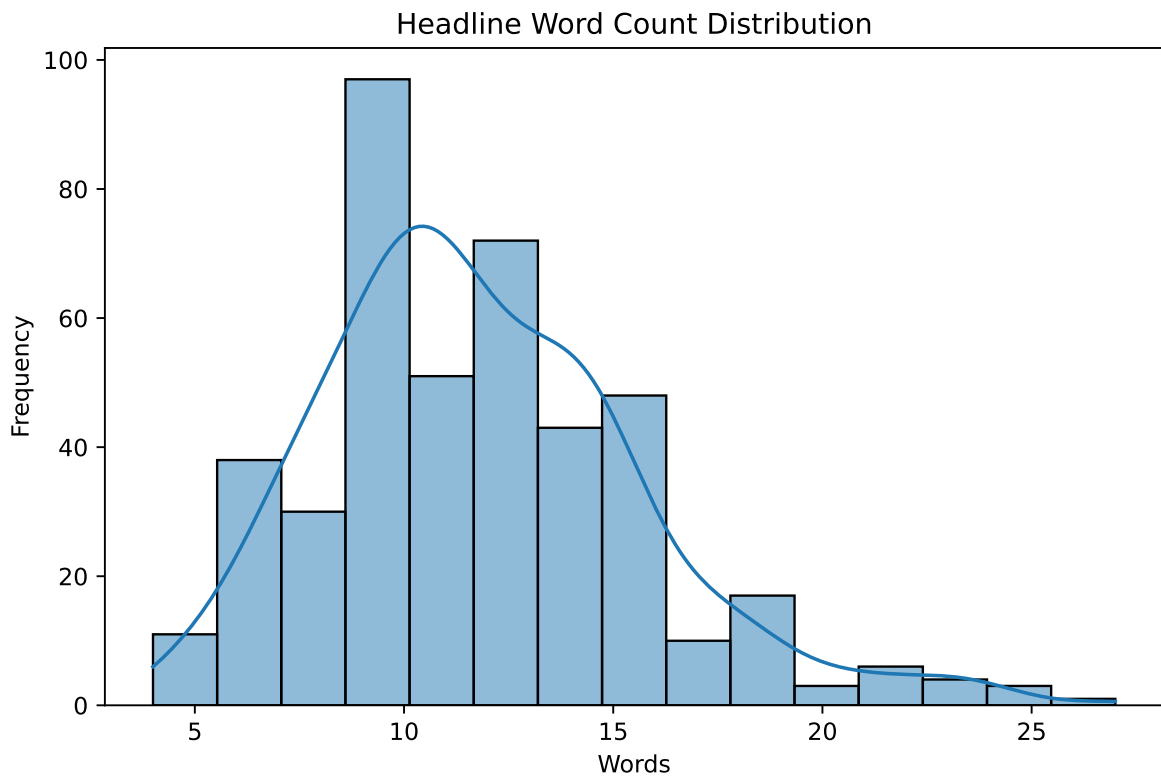
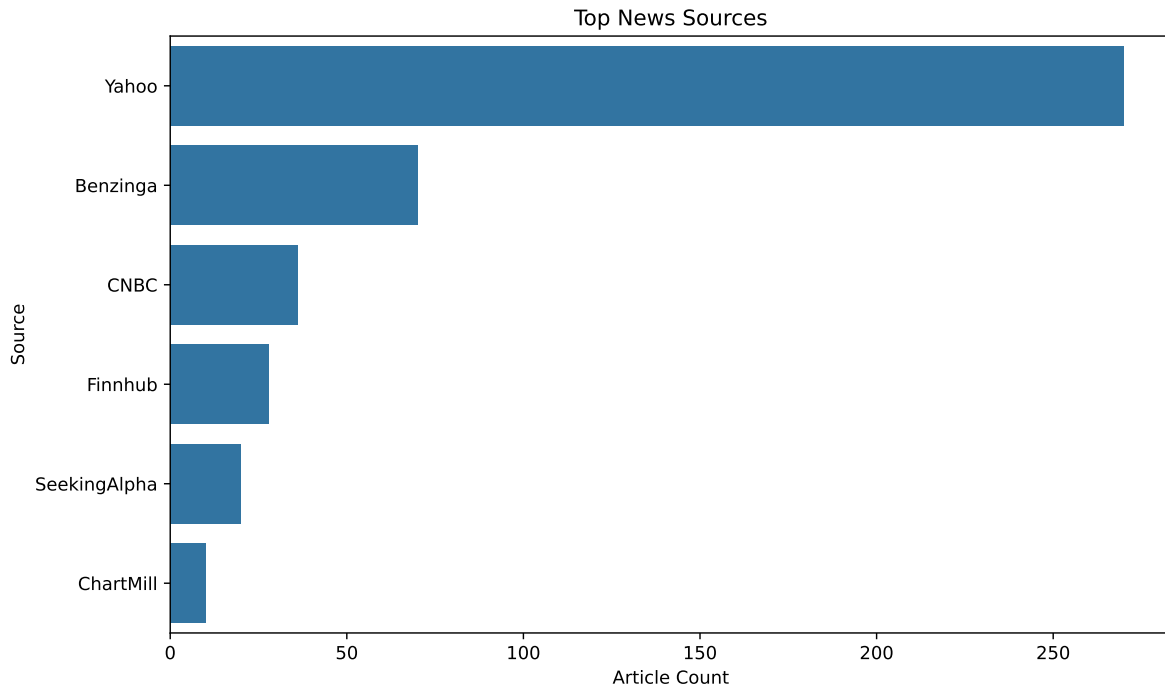
source_sentiment.plot(
    kind="bar",
    stacked=False,
    figsize=(10,6)
)

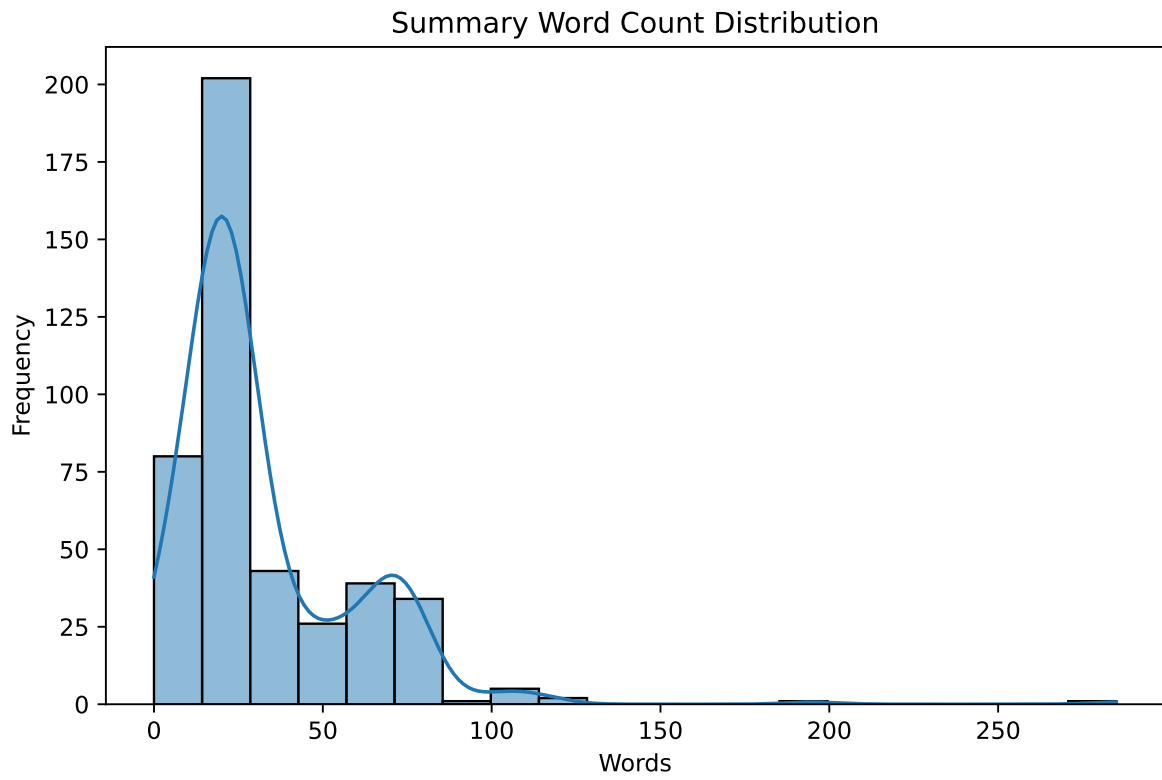
plt.title("Sentiment by News Source")
plt.xlabel("Source")
plt.ylabel("Article Count")
plt.xticks(rotation=45)

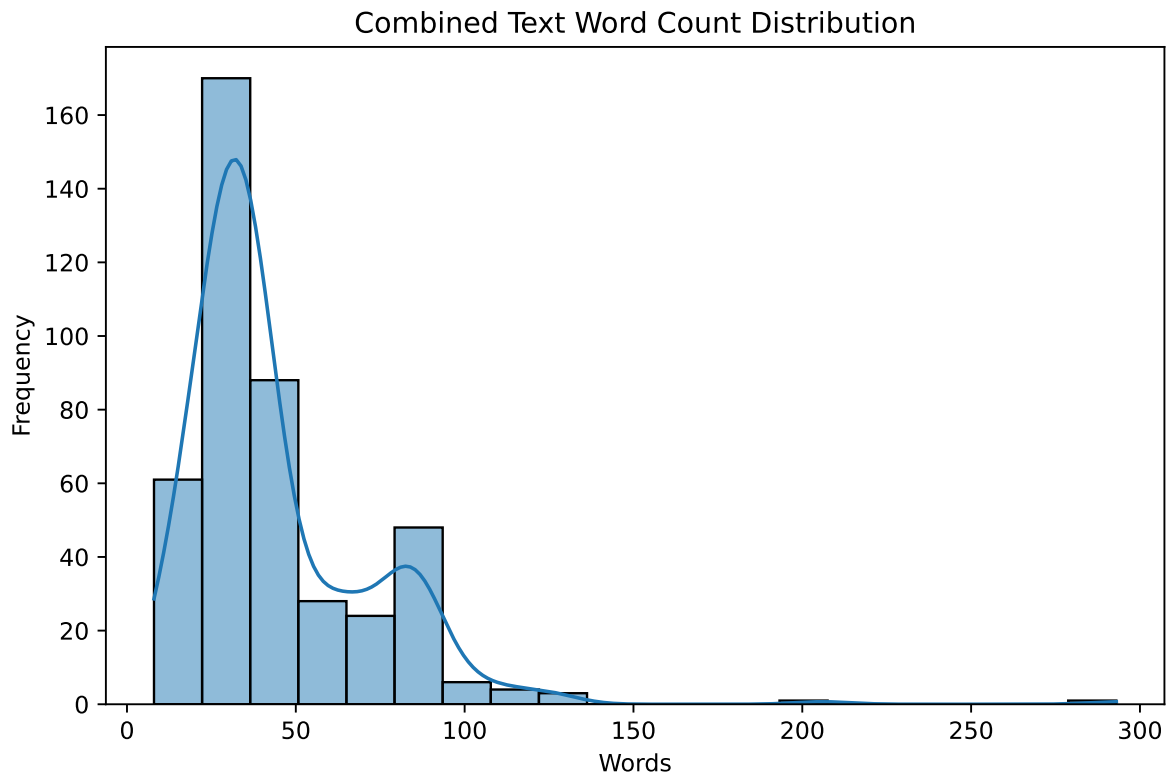
```

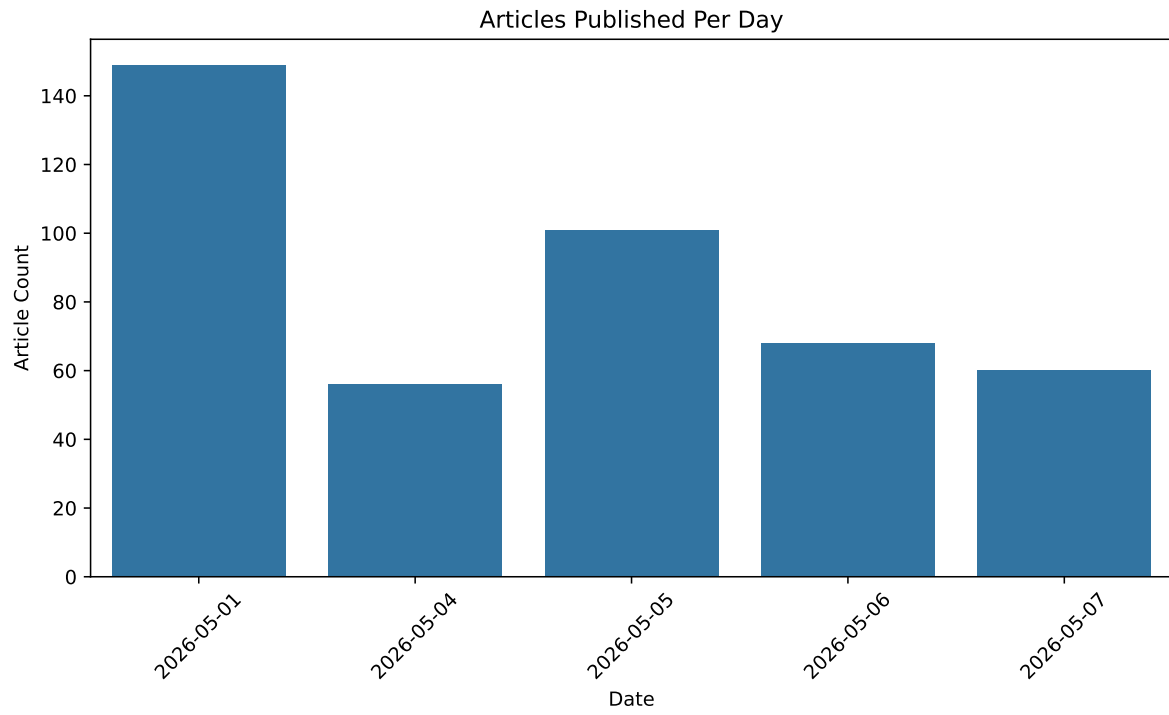


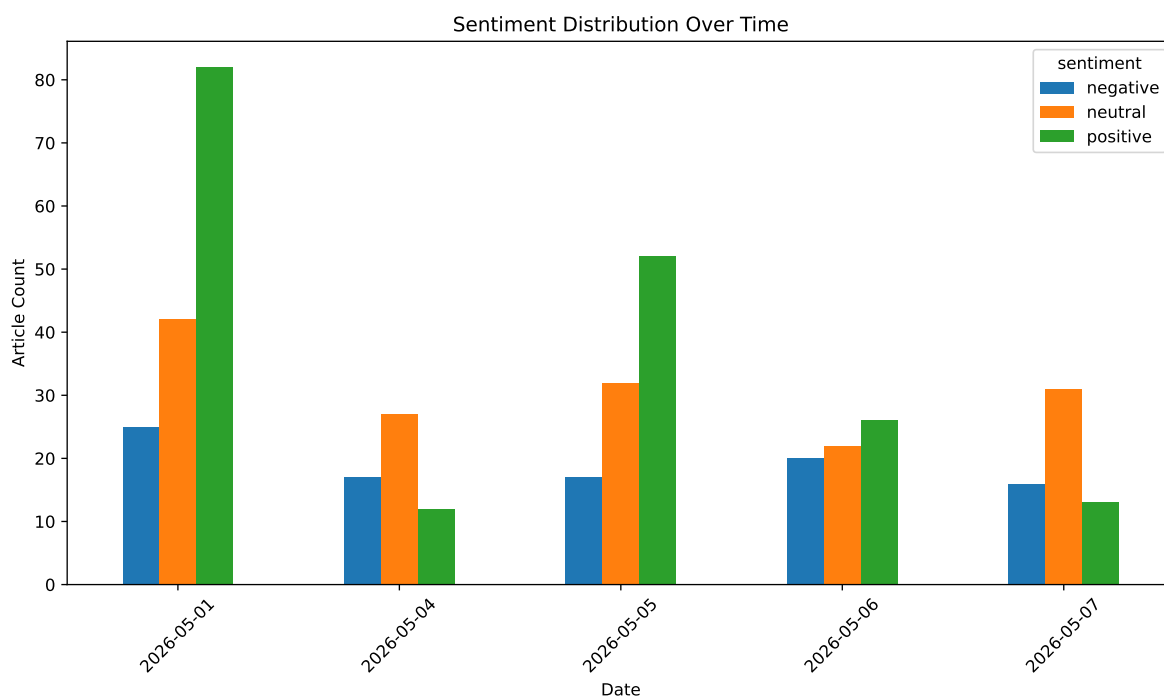
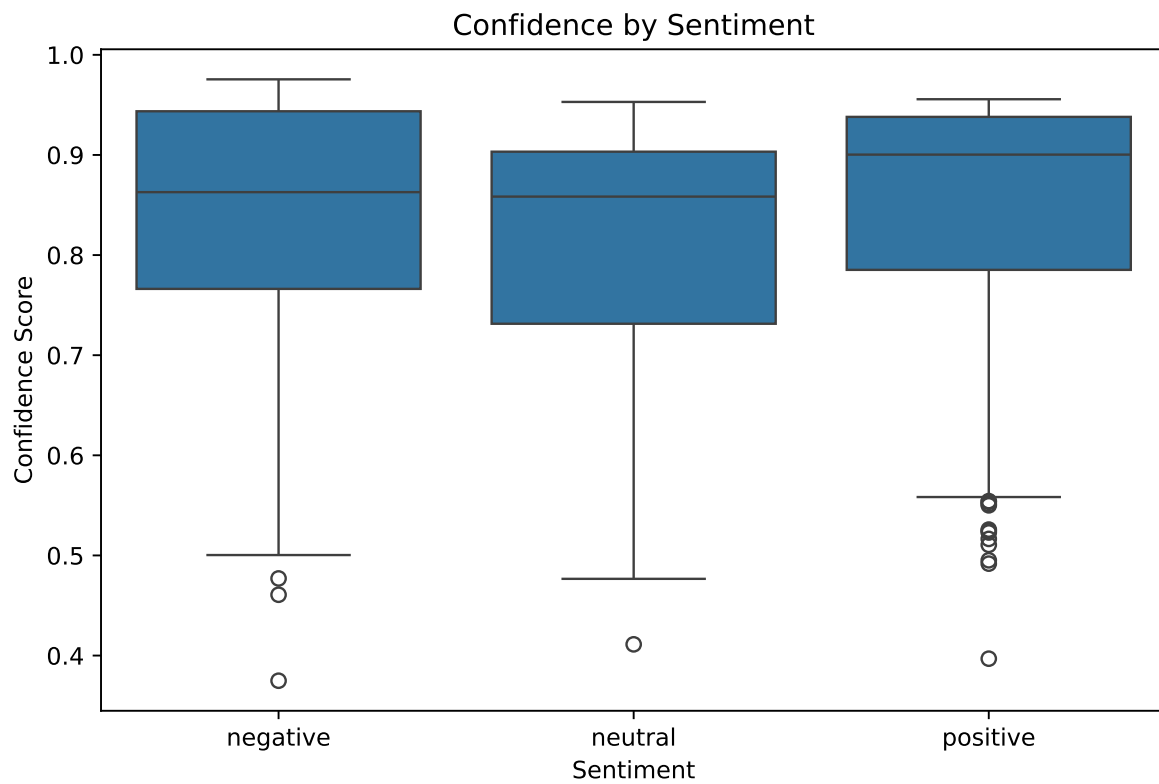




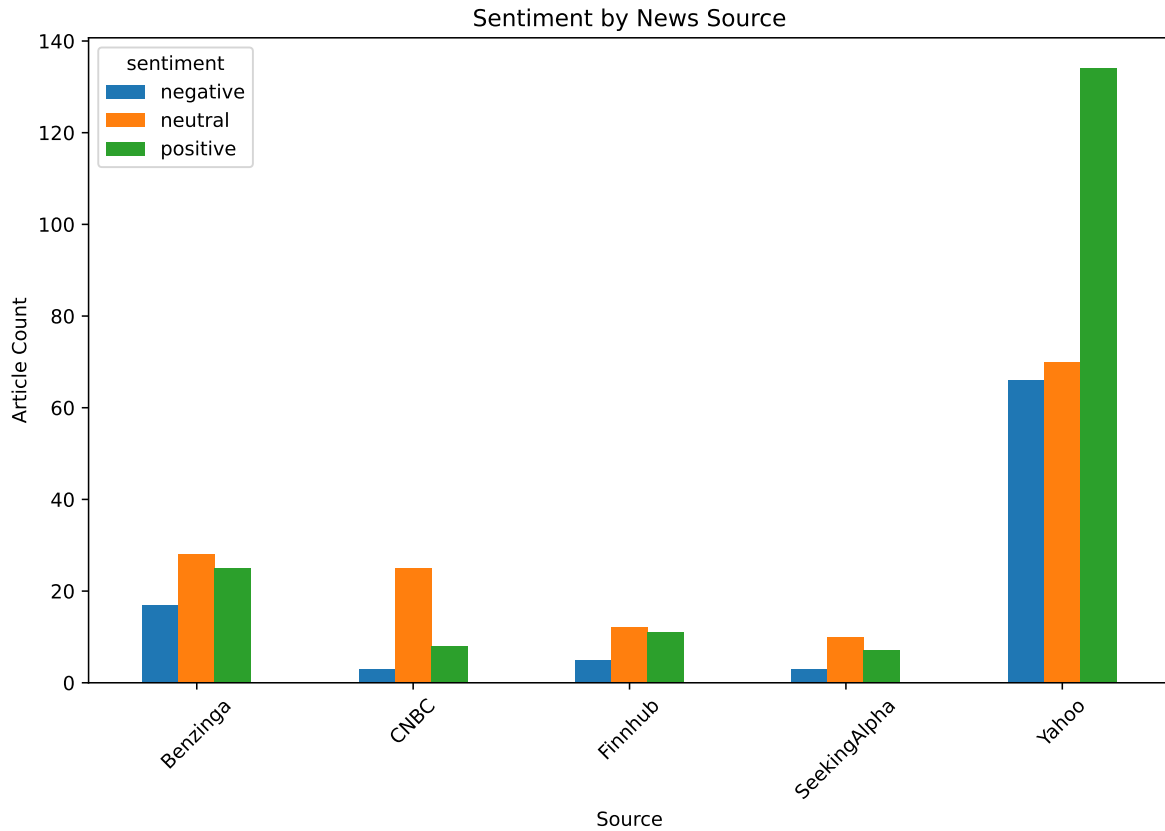








```
(array([0, 1, 2, 3, 4]),
 [Text(0, 0, 'Benzinga'),
  Text(1, 0, 'CNBC'),
  Text(2, 0, 'Finnhub'),
  Text(3, 0, 'SeekingAlpha'),
  Text(4, 0, 'Yahoo')])
```



Expected Distributions

Several variables followed anticipated patterns given the financial nature of the data:

- **Ticker Distribution:** As expected, the Ticker Distribution shows 100% of the 174 articles belong to “AAPL.” This is expected as the extraction script was specifically scoped to this ticker.
- **Articles Published Per Day:** The Articles Published Per Day graph shows a massive spike on 2026-05-01. This aligns with the “Apple Q2 Earnings” headlines noted in the output, as earnings reports typically trigger a high volume of immediate coverage.

- Combined Text Word Count Distribution: The distribution graph is right-skewed, which is typical for news data. Most articles consist of a concise headline and a short summary (averaging ~44 words), while a few outliers provide much more extensive context.
- Sentiment Distribution: The Sentiment Distribution is predominantly positive. Given that the headlines mention Apple “beating estimates” and hitting “record highs,” a positive skew is the logical outcome of a strong earnings report.

Unexpected Distributions and Observations

Some variables revealed more nuanced or surprising trends:

- Distribution of FinBERT Confidence Scores: While one might expect a normal distribution the graph shows a strong left skew. The model is highly confident (>0.90) for a large portion of the data (79 articles). This suggests the language used in these financial reports is very explicit and aligns well with FinBERT’s training data.
- Confidence by Sentiment: The “positive” sentiment boxplot shows many low-confidence outliers. This is unexpected because positive news is usually straightforward; however, it suggests some “positive” articles might contain mixed language (e.g., “Apple Surged... 3 Concerns Lurking Beneath”) that dampens the model’s certainty.
- Sentiment by News Source: The Sentiment by News Source chart reveals significant source bias. Yahoo is overwhelmingly positive, while CNBC appears almost exclusively neutral in its reporting style for this period. This highlights how the choice of data source can fundamentally shift the perceived market sentiment.
- Sentiment Distribution Over Time: The sentiment shifted from neutral/positive on April 30th to overwhelmingly positive on May 1st (Earnings Day), before cooling back toward neutral on May 2nd. The rapid “digestion” of the news by the market is clearly visible here.

Bias-Related EDA Code

```
# BIAS EXPLORATION

print("\n=====")
print("BIAS ANALYSIS")
print("=====")

# Bias check 1: source imbalance
source_counts = df["source"].value_counts()
print("\nSource imbalance:\n", source_counts)

# Bias check 2: sentiment skew
sentiment_counts = df["sentiment"].value_counts(normalize=True)
```

```

print("\nSentiment proportions:\n", sentiment_counts)

# Bias check 3: missing data patterns
print("\nMissing data pattern:\n", df.isna().mean())

# Bias check 4: sentiment by source
print("\nSentiment by source:\n")
print(df.groupby("source")["sentiment"].value_counts())

# Bias check 5: potential overconfidence
high_confidence = df[df["sen_confidence"] > 0.90]
print("\nHigh confidence articles (>0.90):", len(high_confidence))

# DATA PRODUCT AND DESCRIPTION
df.to_csv("stocks_v2.csv", index=False)

# =====
# VOLUME vs SENTIMENT VISUALIZATION
# =====
# Build a per-day frame using `results` (transformed_dataset), which
# already has aligned `volume` and per-article `sentiment_results`.
#
# Daily net sentiment score:
#   for each article: +confidence if positive, -confidence if negative, 0 if neutral
#   daily score = mean of those signed confidences (range roughly [-1, 1])

sentiment_sign = {"positive": 1, "negative": -1, "neutral": 0}

rows = []
for day, payload in results.items():
    scores = [
        sentiment_sign.get(s["label"].lower(), 0) * s["score"]
        for s in payload["sentiment_results"]
    ]
    net_sentiment = sum(scores) / len(scores) if scores else 0.0
    rows.append({
        "date": day,
        "volume": payload["volume"],
        "article_count": len(payload["articles"]),
        "net_sentiment": net_sentiment,
    })

```



```

daily_df = pd.DataFrame(rows)
daily_df["date"] = pd.to_datetime(daily_df["date"])
daily_df = daily_df.sort_values("date").reset_index(drop=True)

print("\nDaily volume vs sentiment:")
print(daily_df)

# Daily volume vs sentiment
fig, ax1 = plt.subplots(figsize=(11, 6))

x_labels = daily_df["date"].dt.strftime("%Y-%m-%d")

ax1.bar(
    x_labels,
    daily_df["volume"],
    color="steelblue",
    alpha=0.6,
    label="Trade Volume",
)
ax1.set_xlabel("Date")
ax1.set_ylabel("Trade Volume", color="steelblue")
ax1.tick_params(axis="y", labelcolor="steelblue")
ax1.tick_params(axis="x", rotation=45)

ax2 = ax1.twinx()
ax2.plot(
    x_labels,
    daily_df["net_sentiment"],
    color="darkorange",
    marker="o",
    linewidth=2,
    label="Net Sentiment",
)
ax2.axhline(0, color="gray", linestyle="--", linewidth=1)
ax2.set_ylabel("Net Sentiment (-1 to +1)", color="darkorange")
ax2.tick_params(axis="y", labelcolor="darkorange")
ax2.set_ylim(-1, 1)

plt.title(f"{symbol}: Daily Trade Volume vs. FinBERT Net Sentiment")
fig.tight_layout()
plt.show()

```

```
=====
BIAS ANALYSIS
=====
```

Source imbalance:

```
source
Yahoo          270
Benzinga       70
CNBC           36
Finnhub        28
SeekingAlpha   20
ChartMill      10
Name: count, dtype: int64
```

Sentiment proportions:

```
sentiment
positive    0.426267
neutral     0.354839
negative    0.218894
Name: proportion, dtype: float64
```

Missing data pattern:

```
ticker          0.0
headline        0.0
summary         0.0
text            0.0
source          0.0
date            0.0
sentiment       0.0
sen_confidence  0.0
headline_word_count  0.0
summary_word_count  0.0
text_word_count  0.0
dtype: float64
```

Sentiment by source:

```
source      sentiment
Benzinga    neutral    28
            positive    25
            negative    17
CNBC        neutral    25
```

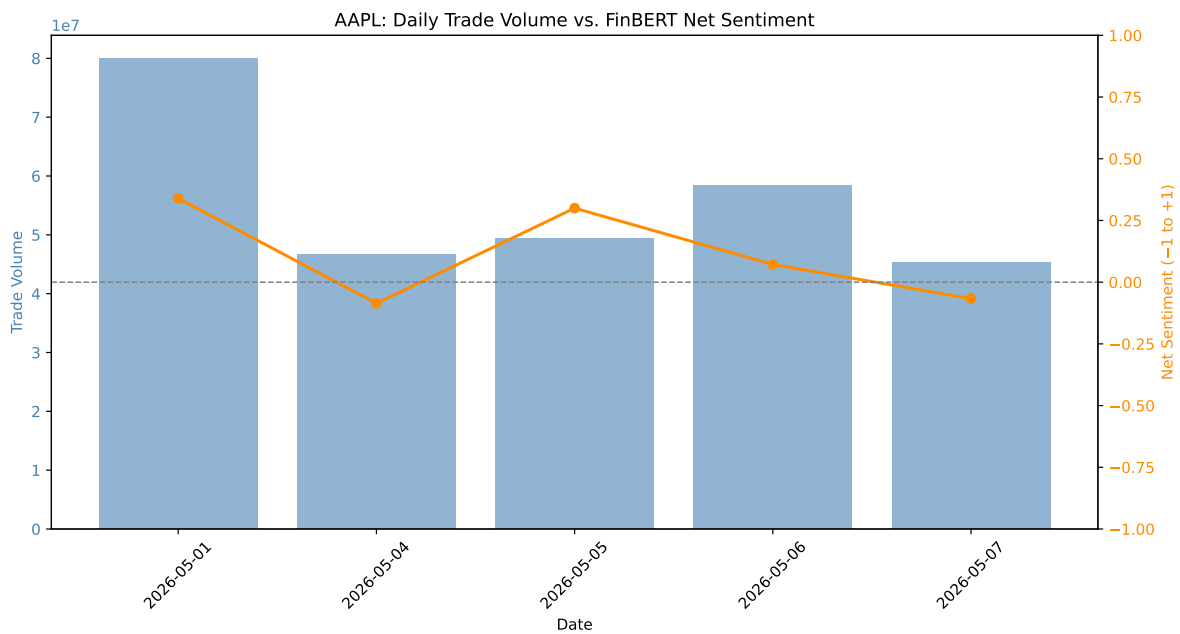
	positive	8
	negative	3
ChartMill	neutral	9
	negative	1
Finnhub	neutral	12
	positive	11
	negative	5
SeekingAlpha	neutral	10
	positive	7
	negative	3
Yahoo	positive	134
	neutral	70
	negative	66

Name: count, dtype: int64

High confidence articles (>0.90): 176

Daily volume vs sentiment:

	date	volume	article_count	net_sentiment
0	2026-05-01	79915400	149	0.339296
1	2026-05-04	46668400	56	-0.085853
2	2026-05-05	49311700	101	0.299959
3	2026-05-06	58336100	68	0.071676
4	2026-05-07	45224300	60	-0.066934



Data Product & Guide

```
## Variable descriptions
```

```
### Quantitative
```

```
sen_confidence (continuous/quantitative) - how certain Finbert is about its sentiment rating
```

```
headline_word_count (discrete/quantitative) - the number of words in the headline
```

```
summary_word_count (discrete/quantitative) - the number of words in the summary
```

```
text_word_count (discrete/quantitative) - the total number of words in the headline and summary
```

```
### Qualitative
```

```
ticker (categorical) - a unique abbreviation for the name of the stock (e.g. AAPL for Apple)
```

```
headline* - the title for the news article; intended to grab the reader's attention and highlight key points
```

```
summary* - text that condenses the main points of the article; generated by Finnhub using NLP
```

```
text* - the headline and summary combined/concatenated with a space in between
```

```
source (categorical) - the news source that the article was published in
```

```
sentiment (categorical/qualitative) - general attitude towards a stock, which can be positive, negative, or neutral
```

```
### Temporal
```

```
date (temporal) - the time the article was published in yyyy-mm-dd format
```

```
* the values for these variables are "free-response"-like or unique
```

```
# to get the variable data types  
df.info()
```

```
<class 'pandas.DataFrame'>
```

```

RangeIndex: 434 entries, 0 to 433
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   ticker                434 non-null    str
1   headline              434 non-null    str
2   summary               434 non-null    str
3   text                  434 non-null    str
4   source                434 non-null    str
5   date                  434 non-null    datetime64[us]
6   sentiment              434 non-null    str
7   sen_confidence        434 non-null    float64
8   headline_word_count   434 non-null    int64
9   summary_word_count    434 non-null    int64
10  text_word_count       434 non-null    int64
dtypes: datetime64[us](1), float64(1), int64(3), str(6)
memory usage: 37.4 KB

```

Code for data product

To export the final DataFrame to a csv file, run `df.to_csv("stocks.csv", index=False)`.

User guide

1. Download the installer for Positron from this [link](#). Make sure to accept the license agreement by checking the box. Choose the file that matches your platform. The platform information is available if you visit System Information for Windows or About This Mac for MacOS.
2. Clone the repository from this [link](#) using `git clone git@github.com:lejo/INST447-Final-Project.git` for SSH.
3. Open the “INST447-Final-Project” folder in Positron by clicking on the Open button with the file icon at the top left of the application.
4. Go to the terminal at the bottom half of the interface. Make sure that you are in the correct folder. Your path should have “INST477-Final-Project.” If not, use the `cd` command to get there. Here is a [tutorial](#) on using the command line.
5. Make a virtual environment by running `python3 -m venv .venv`.
6. Activate the environment using `source .venv/bin/activate` for MacOS/Linux. For Windows Powershell, use `.\venv\Scripts\Activate.ps1`.

7. Install the dependencies from the `requirements.txt` file using `pip install -r requirements.txt`.
- 7.5. Check that `quarto` is installed in the environment using `quarto -v` in the terminal. If it isn't installed, install it using `brew install --cask quarto` for MacOS if you have Homebrew. Otherwise, you can install it from this [website](#). You may have to restart Positron and your terminal.
8. Open the `.qmd` file by clicking on it from the file explorer and render it by clicking the "Preview" button close to the top of the opened file. A side bar on the right should open, and there should be a square with an arrow icon. Clicking the icon will open a new browser window, which will give you access to the notebook code and output running on localhost. Use `CTRL + C` to quit.